Transverse information beyond thermal scalar limits in colliding layers

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(Dated: October 8, 2026)

Thermal scalar observables can approach the same limits while coherent layer measurements retain information about a collapsing geometry. We establish this separation for the actual Green operators of finitely many weighted curves approaching a common curve. A joint separation–temperature limit determines the partition function, normalized heat state, and transverse population, including their high-frequency tails. In a subsequent high-temperature limit, the one-excitation entropy differences vanish, whereas M equally weighted layers remain distinguishable from their collapsed reference with trace distance tending to $1 - 1/M$. The finite crossover has a strict counting deficit, and its low-temperature transverse population detects rotation of the lowest spectral branch. We then distinguish controlled one-excitation preparation from a full number-conserving Bose ensemble. Calibrated occupation thresholds tolerate fixed sorting errors; a common infrared filter yields exponentially small error with an explicit stopband-leakage budget. Exact circular Green blocks give a finite instance with average error below 0.0032 under specified collection loss, sorting errors, and dark counts. Preparation, occupied modes, and the common-vacuum assumption are also included.

I. INTRODUCTION

Thermal measurements compress a spectrum. Partition functions, mean energies, and entropies summarize the occupations of its modes, but a measurement that mixes those modes also probes their orientation in the accessible Hilbert space. The distinction is particularly consequential when a geometric limit and a thermal limit occur together. Modes that disappear at every fixed spectral index can still carry a finite fraction of the information in a temperature-dependent spectral window. The question addressed here is whether this mechanism survives for an actual continuum geometry, with a controlled measurement and preparation cost.

Consider mass supported on several close curves in a planar carrier. Their common displacement and relative displacements define longitudinal and transverse layer combinations. At fixed energy the curves merge, but wavelengths comparable with their separation resolve the layers. We compute the resulting joint limit rather than first replacing the layers by a single curve. The limiting object is a positive matrix-valued symbol whose off-diagonal entries are decaying exponentials of the normal separations. Its eigenvalues determine thermal scalar quantities; its eigenvectors determine the coherent transverse response.

There are close precedents on both sides of this problem. Singular interactions supported by curves, parallel lines, and colliding supports have a substantial spectral theory [1, 2]. The sharp one-dimensional Lieb–Thirring inequality for measures [3] controls a finite-profile counting deficit that appears below. Neither singular interactions nor their exponential Birman–Schwinger kernels are new ingredients. Concentrating physical mass has also been studied through boundary layers and thin heavy inclusions [4, 5]; their bulk-mass and boundary regimes differ from the interior pushforward probability masses used here. High-index spectral limits under concentrated mass have also been studied for branched surfaces with Kirchhoff junctions [6]. Our limit instead concerns fixed-total-mass curves that collide and their trace-class thermal information laws. Matrix semiclassical trace formulas and nonsmooth matrix-symbol tests provide established analytic context [7–9]. The singular inverse symbol and both frequency tails still require the actual-operator comparison proved below. On the measurement side, optical source discrimination already uses a reference mode and its orthogonal complement to detect information missed by direct imaging [10]. The failure of an ideal dark-port rule under imperfect mode sorting, and its replacement by a calibrated threshold, are also established [11]. Background-robust discrimination using reduced spatial-mode demultiplexing has recently been studied in a weak-source Poisson regime [12]; the count bounds below retain full Bose occupation statistics. Bosonic determinant identities and passive optical synthesis provide the relevant counting and implementation tools [13, 14].

The contribution is the connection of these ingredients through a specific continuum limit and its operational consequences. We prove an all-mode joint law for the actual curved Green operators, including a trace-norm comparison on a common space. This gives a separation between thermal scalar agreement and transverse discrimination, a strict finite-crossover deficit, and a profile-dependent distinction between excited-branch occupation and rotation of the lowest branch. We then derive preparation and measurement statements in two explicitly different ensembles. In the full Bose model a common low-frequency rejection filter preserves a macroscopic signal while controlling the largest modal fluctuations; its error exponent has an explicit dependence on stopband leakage. Finally, a finite circular example replaces an unspecified asymptotic onset by exact Green matrices and validated probability bounds.

The physical model is specified at two levels. The continuum operator describes a static Dirichlet energy with mass on curves. Its quantum access model uses

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common, number-conserving modes, which can be programmed in a finite optical encoding. An ordinary mechanical quantization can require an additional Bogoliubov transformation because its vacuum changes with the frequencies. Separating these levels allows the statistical conclusions to be stated without attributing an unbuilt measurement architecture to a membrane. The distinction between thermal scalars and coherence also has antecedents in quantum thermodynamics [15]. Here each state is Gibbs-diagonal in its own Hamiltonian: the comparison concerns different geometric Hamiltonians in a common layer alignment, not coherence relative to one fixed Hamiltonian or an additional work-extraction law. Full derivations are included in the appendices.

II. GEOMETRY, ALIGNMENT, AND THE RESOLVED THERMAL SCALE

Let Γ be a smooth embedded closed curve of length L in a bounded planar Dirichlet carrier Ω. Its fixed tubular neighborhood lies strictly inside Ω. Write s for arclength, $\gamma_0(s)$ for its position, and $n(s)$ for a chosen unit normal. Fix distinct heights $a_1 < \cdots < a_M$, positive weights α_i with $\sum_i \alpha_i = 1$, and a smooth positive density w satisfying $\int_\Gamma w\,\mathrm{d}s = 1$. For sufficiently small $h > 0$, the mass measure is

$$\nu_h = \sum_{i=1}^{M} \alpha_i (s \mapsto \gamma_0(s) + h a_i n(s))_* (w(s)\,\mathrm{d}s). \tag{1}$$

Thus the mass is prescribed by pushforward, not by inserting the arclength Jacobian of each offset curve. Fixed absorbing sets outside the tube may be incorporated into the carrier.

Restrict the Dirichlet Green kernel to these curves and conjugate by $\sqrt{\alpha_i w}$. The resulting positive compact inverse R_h acts on the common space

$$\begin{aligned} \mathcal{H} &= L^2(\Gamma, \mathrm{d}s) \otimes \mathbb{C}^M, \\ v_i &= \sqrt{\alpha_i}, \\ P &= vv^*, \\ Q &= I - P. \end{aligned} \tag{2}$$

Equivalently, $L_h = R_h^{-1}$ is the energy operator obtained by minimizing the bulk Dirichlet energy at fixed curve values, with the stated mass normalization. The collapsed inverse is $R_0 = R_\mathrm{b} \otimes P$. Its Q complement has zero heat weight: it represents excluded modes, not additional zero-energy excitations.

Fix a dispersion exponent $\gamma > 0$ and set

$$\begin{aligned} H_h &= L_h^\gamma, \\ \beta &= \tau h^\gamma, \\ A_h &= \mathrm{e}^{-\beta H_h}, \\ \rho_h &= A_h / Z_h, \\ Z_h &= \operatorname{Tr} A_h. \end{aligned} \tag{3}$$

Here and below the displayed energy and temperature scales are dimensionless. The mechanical frequency convention corresponds to $\gamma = 1/2$ when L_h is a squared-frequency operator. A reference state $\rho_{0,h}$ uses R_0 at the same β. Every geometric parameter, M, the positive weights, γ, and τ is fixed before $h \downarrow 0$. Limits in τ are subsequent limits; none of the conclusions presupposes uniformity when the number of layers grows or their weights degenerate with h.

For $q = |\xi| > 0$ define

$$\begin{aligned} T(q)_{ij} &= \sqrt{\alpha_i \alpha_j}\, \mathrm{e}^{-q|a_i - a_j|}, \\ B(s,\xi) &= \frac{w(s)}{2q} T(q), & B_0(s,\xi) &= \frac{w(s)}{2q} P, \\ F(s,\xi) &= \mathrm{e}^{-\tau B(s,\xi)^{-\gamma}}, & F_0(s,\xi) &= \mathrm{e}^{-\tau (2q/w(s))^\gamma} P. \end{aligned} \tag{4}$$

The value of the heat function at a zero inverse eigenvalue is zero. The exponential matrix $T(q)$ is strictly positive for distinct heights. It contains the entire finite-profile crossover; the remote boundary contributes lower-order terms on this scale.

III. WHAT THERMAL SCALAR MEASUREMENTS MISS

Define the phase-space coefficients

$$\begin{aligned} C(\tau) &= \frac{1}{2\pi} \int_\Gamma \int_{\mathbb{R}} \operatorname{tr} F \,\mathrm{d}\xi\,\mathrm{d}s, \\ C_Q(\tau) &= \frac{1}{2\pi} \int_\Gamma \int_{\mathbb{R}} \operatorname{tr}(QF) \,\mathrm{d}\xi\,\mathrm{d}s, \\ C_0(\tau) &= \frac{\Gamma(1 + 1/\gamma)}{2\pi \tau^{1/\gamma}}. \end{aligned} \tag{5}$$

The actual operators obey

$$h Z_h \longrightarrow C(\tau), \tag{6}$$

$$h Z_{0,h} \longrightarrow C_0(\tau), \tag{7}$$

$$\operatorname{Tr}(Q\rho_h) \longrightarrow p_Q(\tau) = C_Q(\tau)/C(\tau), \tag{8}$$

$$\frac{1}{2} \left\| \rho_h - \rho_{0,h} \right\|_1 \longrightarrow \tag{9}$$

$$D(\tau) = \frac{1}{4\pi} \int_\Gamma \int_{\mathbb{R}} \operatorname{tr} \left| \frac{F}{C(\tau)} - \frac{F_0}{C_0(\tau)} \right| \mathrm{d}\xi\,\mathrm{d}s. \tag{10}$$

For a nontrivial profile, $0 < p_Q(\tau) \le D(\tau) < 1$ at every fixed $\tau > 0$. These statements include the low- and high-frequency tails of the actual Green operators, not only a formal symbol calculation. Appendix A gives the positive Fourier comparison and trace argument.

The clearest separation occurs upon subsequently taking $\tau \downarrow 0$. Then $C/C_0 \to 1$, while

$$p_Q(\tau) \longrightarrow 1 - \sum_i \alpha_i^2. \tag{11}$$

For equal masses, the complete trace-distance limit agrees with this simple layer measurement. The usual binary discrimination relation [16] then gives its operational meaning:

$$\lim_{\tau\downarrow 0}\lim_{h\downarrow 0}\frac{1}{2}\left\|\rho_h-\rho_{0,h}\right\|_1=1-\frac{1}{M}. \tag{12}$$

With equal prior probabilities, a single controlled excitation can therefore be classified with limiting error $1/(2M)$. On N independent controlled excitations, declaring the split geometry when at least one Q event occurs gives limiting error $1/(2M^N)$. The latter is the performance of this specified receiver, without a claim of optimality over all collective N-copy measurements.

This surviving information is compatible with agreement of a substantial set of thermal scalars. For $p>0$ put

$$C_p(\tau)=\frac{1}{2\pi}\int \mathrm{tr}\big[(\tau B^{-\gamma})^p \mathrm{e}^{-\tau B^{-\gamma}}\big]\,\mathrm{d}s\,\mathrm{d}\xi. \tag{13}$$

At fixed τ, the one-excitation entropy and energy satisfy

$$\begin{aligned} S(\rho_h)&=\log(1/h)+\log C+C_1/C+o(1),\\ \beta\langle H_h\rangle&\longrightarrow C_1/C,\\ \beta^2\,\mathrm{Var}_{\rho_h}(H_h)&\longrightarrow C_2/C-(C_1/C)^2. \end{aligned} \tag{14}$$

Every fixed positive-order Rényi entropy has an analogous formula. As $\tau\downarrow 0$, the additive entropy differences from the collapsed reference vanish, and $C_p/C\to\Gamma(p+1/\gamma)/\Gamma(1/\gamma)$. Both states also have the same limiting expectations against fixed smooth spatial multipliers. Thermal scalar agreement here means these specified limits. It does not mean equality of the spectra or of arbitrary measurements at finite separation and temperature.

IV. COUNTING DEFICIT AND ROTATION OF THE LOWEST BRANCH

The same matrix symbol connects the crossover with a classical one-dimensional spectral problem. Let $-\kappa_j(g)^2$ be the negative eigenvalues of

$$-\frac{\mathrm{d}^2}{\mathrm{d}a^2}-g\sum_i \alpha_i\delta_{a_i}\quad\text{on }\mathbb{R}, \tag{15}$$

with absent branches contributing zero. The actual growing-index count is

$$\begin{aligned} h\,\#\{\lambda_k(L_h)\le E/h\}&\longrightarrow\\ \mathcal{N}(E)=\frac{1}{\pi}\int_\Gamma\sum_j\kappa_j(w(s)E)\,\mathrm{d}s&<\frac{E}{2\pi} \end{aligned} \tag{16}$$

for $E>0$ and at least two distinct positive-mass layers. The strict inequality is the finite-profile consequence of the sharp Lieb–Thirring bound for measures [3]; we include its finite-atomic argument to make the connection transparent. Integrating the counting law gives $C(\tau)<C_0(\tau)$ at every finite τ. Thus the high-temperature agreement of leading coefficients coexists with a strict crossover deficit, not with an exactly invisible spectral change.

There is also a useful distinction within the transverse response. Define

$$d_i=\sum_j\alpha_j|a_i-a_j|, \tag{17}$$

$$G=\sum_i\alpha_i d_i, \tag{18}$$

$$V=\sum_i\alpha_i(d_i-G)^2,$$

$$I_k=\int_\Gamma w(s)^k\,\mathrm{d}s, \tag{19}$$

$$G_\gamma=\Gamma(1+1/\gamma). \tag{20}$$

After the joint limit, the large-τ expansions are

$$\begin{aligned} C(\tau)&=\frac{G_\gamma}{2\pi}\tau^{-1/\gamma}-\frac{GI_2\Gamma(1+2/\gamma)}{4\pi}\tau^{-2/\gamma}+O(\tau^{-3/\gamma}),\\ p_Q(\tau)&=\frac{VI_3\Gamma(1+3/\gamma)}{12G_\gamma}\tau^{-2/\gamma}+O(\tau^{-3/\gamma}). \end{aligned} \tag{21}$$

Excited branches of the limiting symbol are exponentially suppressed in τ, but the lowest eigenvector rotates away from the fixed common layer vector v. The mean-distance statistic G in the first line is the classical weak-coupling coefficient [17, 18]. The second line measures the rotation. Its coefficient vanishes only for a single layer or two equally weighted layers; in the latter case the lowest eigenvector remains exactly symmetric and the transverse response decays exponentially. Replacing Q by the instantaneous excited-branch projector would therefore remove a genuine observable effect. The tridiagonal inverse of $T(q)$ gives all finite branch thresholds and proves these assertions in Appendix B.

V. PREPARING THE ONE-EXCITATION STATE

The state ρ_h is a normalized one-excitation Gibbs state. Conditioning a full equilibrium sample on exactly one excitation need not be an efficient way to prepare it. A uniform spectral estimate for the actual curves resolves the finite-dimensional requirement:

$$r_j(R_h)\le K/j,\qquad \lambda_j(L_h)\ge cj. \tag{22}$$

The constants are uniform through the collision at fixed geometric data. Self-layer regularity and positivity of the full block operator prove this without assuming uniformly smooth cross-layer kernels. For fixed trace accuracy δ, the required energy truncation has

$$J=O\big(h^{-1}[1+\log(1/\delta)]^{1/\gamma}\big). \tag{23}$$

A common tensor space containing the retained vectors of both hypotheses has dimension at most $M(M+1)J$ and preserves P and Q exactly. Selected energy and entropy observables are controlled by their own weighted tails, rather than inferred from trace distance alone.

On this known finite encoding, start from the maximally mixed one-excitation input and apply the retained heat contraction $A_J^{1/2}$. The success probability is Z_J/D, bounded below at fixed accuracy because both Z and D scale as h^{-1}. An optical unitary dilation realizes the contraction; universal interferometer synthesis [14] costs $O(D^2)$ elements and depth $O(D)$. These are costs of a programmed state source with known matrix data. Input heralding, rejected trials, mode-dependent loss, and reset remain part of its resource budget. Balanced direct eigenvector preparation offers a different tradeoff, described in Appendix D.

For $\gamma = 1/2$, subordination gives an exact alternative using a no-loss evolution generated by L:

$$\mathrm{e}^{-\beta\sqrt{L}} = \int_0^\infty \frac{\beta \mathrm{e}^{-\beta^2/(4u)}}{2\sqrt{\pi}u^{3/2}} \mathrm{e}^{-uL}\,\mathrm{d}u. \tag{24}$$

The successful, unnormalized outputs must be pooled before normalization. The sampling law has infinite mean time, but a controlled cutoff gives fixed accuracy; alternatively, aborting at the first detected loss gives mean attempted evolution time $O(h)$ in the dimensionless generator units. Bounding the physical decay rate rescales this time to $O(1)$ at fixed accuracy. The construction thus pays for the growing generator bandwidth instead of interpreting a shrinking mathematical time as unlimited laboratory speed. A local bulk stiffness matrix reduces to a generally dense boundary operator, so its static Schur complement alone does not implement the required dynamics.

VI. A FULL BOSE ENSEMBLE IS A DIFFERENT RESOURCE

Now specify a common number-conserving Fock model on $\mathcal{F}_s(\mathcal{H})$:

$$\begin{aligned} \mathbb{H}_h &= \mathrm{d}\Gamma(H_h), \\ \varrho_h &= \det(I - A_h)\,\Gamma(A_h), \\ N_h &= A_h(I - A_h)^{-1}. \end{aligned} \tag{25}$$

The chemical potential is zero and the reference has its thermal state on P and exact vacuum on Q. This ensemble has variable particle number; it is not the canonical ensemble at a fixed number of excitations. Energies are excitation energies relative to the declared vacuum.

For an orthogonal detected subspace E, efficiency $0 \le \eta \le 1$, and $0 \le z \le 1$, the occupation generating function is

$$\mathbb{E}z^K = \det\bigl(I + \eta(1-z)EN_hE\bigr)^{-1}. \tag{26}$$

This familiar bosonic determinant [13, 19] does not require E to commute with A_h. In particular, an ideal Q receiver has zero false-positive probability and

$$p_{\mathrm{off}} \le \exp\{-\log(1+\eta)[C_Q(\tau) + o(1)]/h\}. \tag{27}$$

This is distinguishability of an entire many-excitation ensemble. The same determinant mechanism underlies established thermal source-discrimination receivers [10].

The total resource follows from the continuum law. Define

$$\begin{aligned} \mathcal{L}(\tau) &= \sum_{k\ge1} C(k\tau)/k, & \mathcal{E}(\tau) &= \sum_{k\ge1} C_1(k\tau)/k, \\ \mathcal{V}(\tau) &= \sum_{k\ge1} C_2(k\tau)/k, & \mathcal{B}(\tau) &= \sum_{k\ge1} C(k\tau). \end{aligned} \tag{28}$$

For every $\gamma > 0$, $h\log\mathcal{Z}_B \to \mathcal{L}$, $h\beta\langle\mathbb{H}_h\rangle \to \mathcal{E}$, and $h\beta^2\,\mathrm{Var}(\mathbb{H}_h) \to \mathcal{V}$. The Fock entropy satisfies $hS(\varrho_h) \to \mathcal{L} + \mathcal{E}$. For $0 < \gamma < 1$, the number sum also converges and $h\,\mathrm{Tr}\,N_h \to \mathcal{B}$. At $\gamma = 1/2$, mean particle number is $\Theta(h^{-1})$ and mean excitation energy is $\Theta(h^{-3/2})$. Agreement in a subsequent high-temperature limit concerns ratios of these extensive coefficients, not vanishing additive Fock-entropy differences. Indeed the strict finite-τ partition deficit implies an exponentially small ratio of the split and collapsed Bose partition functions.

The probability that a full Bose sample contains exactly one excitation is $Z_h/\mathcal{Z}_B \le Z_h\mathrm{e}^{-Z_h}$, exponentially small in h^{-1}. Controlled preparation avoids that particular postselection cost; unconditioned Bose readout instead uses the many available excitations. These are two consistent experiments, not interchangeable copy counts.

VII. ROBUST COUNTING AND THE ROLE OF INFRARED REJECTION

A dark-port rule is sensitive to a false-click floor. With a sorter V and reference covariance supported on P, leakage amplitude $\theta = \|QVP\|$ gives reference population at most $\theta^2\,\mathrm{Tr}\,N_{0,h}$. Thus fixed crosstalk can make an any-click rule unreliable as the number of occupied modes grows. This is the familiar reason to replace a dark-port decision by a calibrated count threshold [11].

For the present geometry, let $0 < \gamma < 1$ and let a common detection effect differ from Q by a sufficiently small fixed operator norm. The two mean counts still differ by a positive multiple of h^{-1}. The uniform estimate (22) bounds the largest Bose occupation by $O(h^{-\gamma})$. A geometric-count moment-generating bound then gives a midpoint-threshold error

$$P_{\mathrm{err}} \le \exp[-ch^{\gamma-1}]. \tag{29}$$

Independent calibrated Poisson background with mean $O(h^{-1})$ is allowed. The detector resolves total occupation, with a dynamic range of order h^{-1}; a single bucket-detector bit is a different measurement.

The loss in the exponent comes from very highly occupied infrared modes. A common filter can remove them under both hypotheses. Choose a small fixed $b > 0$ and exclude a P-invariant span containing both low-energy spaces below b/h. Its rank is $O(b/h)$ and its lost mean population is at most $O(b^{1-\gamma}/h)$. For small b, a positive transverse mean gap remains. A subsequent common finite Fourier restriction produces a projection Ξ_h of rank $O(h^{-1})$, commuting with P, on which both covariances are uniformly bounded. A fixed sufficiently small error within this retained space therefore permits

$$P_{\mathrm{err}} \le \exp(-c/h). \tag{30}$$

The filter is constructed from the two known candidate operators and is used unchanged under both hypotheses.

This conclusion has a quantitative stopband condition. For a physical passive contraction T_h into $O(h^{-1})$ output modes, set $\ell_h = \|T_h(I - \Xi_h)\|$ and assume $\|\Xi_h T_h^* T_h \Xi_h - \Xi_h Q\| \le e$. For fixed positive efficiency, if $e + 2\ell_h + \ell_h^2$ stays below the fixed tolerance preserving the positive mean gap, then

$$P_{\mathrm{err}} \le \exp\left[-\frac{c}{h(1+\ell_h^2 h^{-\gamma})}\right]. \tag{31}$$

In particular, amplitude leakage $O(h^{\gamma/2})$ retains the full exponent; at $\gamma = 1/2$ this is $O(h^{1/4})$. The distinction between fixed errors inside the passband and leakage from a highly populated stopband is essential. Appendix E proves the finite-space construction, count concentration, and leakage estimate.

VIII. AN EXACT FINITE CIRCULAR INSTANCE

A rotationally invariant example provides finite numbers without a Weyl approximation. In the unit Dirichlet disk take circles of radii 99/200 and 101/200, each carrying mass 1/2 uniformly in angle. The reference is the mass-one circle of radius 1/2, aligned with the symmetric layer mode. For angular order $n \ge 1$ the exact inverse block is

$$(R_n)_{ij} = \frac{1}{8\pi n}\left[\left(\frac{\min(r_i, r_j)}{\max(r_i, r_j)}\right)^n - (r_i r_j)^n\right]. \tag{32}$$

Both angular signs are included. Use $\gamma = 1/2$, $\beta = 1/20$, and measure the antisymmetric-like output in $50 \le |n| \le 500$. Allow a separate fixed real sorting-angle error in $[-1/10, 1/10]$ radians in each sector, collection efficiency 7/10, and independent Poisson dark counts of total mean at most 1/10. Declare the split geometry when the total recorded occupation is at least five. Validated arithmetic gives the bounds in Table I. There are 902 measured output modes and 1804 prepared-band modes. Mean prepared occupations are approximately 61.7931 and 81.3157 for the split and reference states. The corresponding excitation energies are approximately 3550.20 and 3590.81 in the stated units. The bound is uniform over the specified sectorwise angles and dark-count means. Its derivation uses exact 2×2 functional calculus, geometric occupation generating functions, and interval-certified Chernoff bounds; the complete calculation is in Appendix F 1.

The entire two-layer angular band is a reducing subspace. Its Gaussian marginal may be prepared alone while reproducing these measured statistics; omitted modes of a full ensemble are traced out. The finite instance assumes ideal rejection outside the selected angular band. It establishes neither matched brightness nor finite-parameter thermal scalar agreement. The asymptotic leakage law in Sec. VII is a separate result, not an uncomputed finite tolerance for this example.

IX. PHYSICAL REALIZATION AND INTERPRETATION

Two implementation facts determine how to interpret these results. First, a finite gauge-invariant Gaussian state with covariance N has a positive coherent-state representation: sample a complex Gaussian amplitude with covariance N and prepare that coherent amplitude. This established representation [10] gives a programmed optical implementation of any declared finite marginal. For a general detector subspace the correct input is its compressed covariance, not the Bose function of a compressed Hamiltonian. Dense covariance factorization and sampling cost $O(D^3)$ once and $O(D^2)$ per draw, respectively, with D mode controls. This simulator knows the covariance; an unknown-specimen sensor additionally needs a physical coupling that supplies it.

Second, a common one-particle alignment does not automatically align mechanical vacua [20]. In fixed annihilation coordinates a real symmetric positive matrix H gives the number-conserving Hamiltonian

$$\mathrm{d}\Gamma(H) = \tfrac{1}{2}(x^{\mathsf{T}} H x + p^{\mathsf{T}} H p - \operatorname{Tr} H). \tag{33}$$

An ordinary oscillator instead has $(p^2 + \omega^2 x^2)/2$ and frequency-dependent annihilation operators. Relative to a reference frequency, its vacuum is squeezed, with $r = \frac{1}{2}\log(\omega/\omega_{\mathrm{ref}})$ and mean reference occupation $\sinh^2 r$. Raising a mechanical frequency is therefore not the same operation as making a fixed optical port empty. A native realization must pay for canonical conversion, added

TABLE I. Rigorous bounds for the finite circular count experiment. They bound the specified receiver error, not the optimal error.

Quantity	Upper bound
False positive	0.0057
False negative	0.00072
Equal-prior error	0.0032

noise, bandwidth, and reset, as in the broader optomechanical setting [21]. For example, restoring a fixed frequency unit Ω_* at $\gamma = 1/2$ gives $\beta = \hbar\Omega_*/(k_B T) = \tau\sqrt{h}$. The joint regime then has $T \propto h^{-1/2}$ and a retained frequency band of order $\Omega_* h^{-1/2}$ at fixed truncation accuracy. An optical simulator at a fixed carrier reproduces the programmed occupation statistics, not these mechanical temperature and energy units.

The mathematical infinite-energy reference has controlled finite replacements. In a one-excitation model, retain longitudinal dispersion and add a gap Δ to each of the $M-1$ transverse channels. Their total probability is

$$\frac{(M-1)\mathrm{e}^{-\beta\Delta}}{1+(M-1)\mathrm{e}^{-\beta\Delta}}. \tag{34}$$

For a full Bose reference with D_Q detected transverse modes, their expected detected occupation is bounded by $\eta D_Q/(\mathrm{e}^{\beta\Delta}-1)$. The condition $\beta\Delta - \log D_Q \to +\infty$ suffices for an any-click vacuum approximation, whereas a calibrated count threshold only requires an adequate mean margin. A scalar gap alone on infinitely many modes has infinite thermal trace; longitudinal confinement must remain present.

A collapsing geometry can retain transverse information after a broad set of normalized thermal summaries has converged. Accessing it can use a controlled excitation or a many-excitation ensemble, with different costs. In the latter, the largest modal occupations, rather than the total signal alone, govern robustness. Common infrared rejection separates the signal budget from the fluctuation budget and yields an explicit interpolation between the two error exponents. The finite circular certificate gives a directly reproducible target for a common-mode emulator. A native geometric device would further require a uniform thin-layer, carrier, and mode-conversion approximation over its occupied band.

ACKNOWLEDGMENTS

The author declares the substantial use of consumer-grade Gemini-3 and GPT-6 models for literature search, exploring and developing ideas, codes and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

DATA AVAILABILITY

Numerical results reported here are reproducible from the verification scripts and certificate data in the repository - Repository - github.com/arunchandru/colliding_layers.

Appendix A: Physical mass, alignment, and the complete joint trace law

Let $\Omega \subset \mathbb{R}^2$ be a fixed bounded open set, and let Γ be a smooth embedded closed curve of length ℓ (the quantity denoted by L in the main text). Write $\mathbf{r}(s)$ for its arclength parametrization, with $s \in \mathbb{R}/\ell\mathbb{Z}$, and $n(s)$ for a smooth unit normal. A closed tubular neighborhood of Γ is assumed to lie compactly inside Ω. This allows any fixed absorbing masks a positive distance from the tube, without regularity assumptions on their distant boundaries. Fix a smooth positive function w with $\int_\Gamma w\,\mathrm{d}s = 1$, distinct fixed heights $a_1 < \cdots < a_M$, and positive fixed weights α_i with $\sum_i \alpha_i = 1$. For sufficiently small $h > 0$ put

$$T_{i,h}(s) = \mathbf{r}(s) + h a_i n(s), \qquad \nu_h = \sum_{i=1}^{M} \alpha_i (T_{i,h})_*(w\,\mathrm{d}s). \tag{A1}$$

The sheets are then disjoint and embedded. The measure in (A1) is the physical mass: its expression in the coordinate s is $\alpha_i w(s)\mathrm{d}s$. In particular there is no arclength Jacobian to insert when a sheet is moved.

For clarity, the Green inverse used here has a direct variational definition. Give $H_0^1(\Omega)$ the energy inner product $\int_\Omega \nabla u \cdot \overline{\nabla v}\,\mathrm{d}x$, with no factor $1/2$, and let $J_h u$ be its traces on the sheets, considered as an element of $L^2(\nu_h)$. The usual local trace estimate in a smooth tube, followed by Poincare's inequality on the bounded carrier, makes J_h bounded. The trace lies locally in $H^{1/2}$; compactness of $H^{1/2}(\Gamma) \hookrightarrow L^2(\Gamma)$ makes J_h compact. Its range is dense: smooth data on finitely many disjoint sheets can be extended separately inside disjoint smaller tubes. The physical closed form is

$$\mathcal{E}_h(f) = \inf\left\{ \int_\Omega |\nabla u|^2\,\mathrm{d}x : J_h u = f \right\}, \qquad f \in J_h H_0^1(\Omega).$$

The infimum is attained on the energy-orthogonal complement of $\ker J_h$. That complement is complete in energy norm; the bounded trace map identifies it with this form domain, proving closedness. The inverse of its positive self-adjoint generator is $J_h J_h^*$. Indeed $u = J_h^* f$ is the unique weak solution with sheet source $f\nu_h$, and its trace is $J_h J_h^* f$. Density of the range proves injectivity of this positive compact inverse. Denote the generator by L_h and its inverse by $R_h = L_h^{-1}$.

Let G_Ω be the Dirichlet Green kernel for this energy. The weak solution representation gives the integral kernel $G_\Omega(x,y)$ on $L^2(\nu_h)$. Conjugating by the exact unitary map

$$(U_h f)_i(s) = \sqrt{\alpha_i w(s)}\, f(T_{i,h}(s)) \quad \text{onto} \quad \mathcal{H} = L^2(\Gamma, \mathrm{d}s) \otimes \mathbb{C}^M \tag{A2}$$

gives the matrix kernel $\sqrt{w(s)w(t)\alpha_i\alpha_j}\, G_\Omega(T_{i,h}(s), T_{j,h}(t))$, again denoted

by R_h. All logarithmic diagonal singularities are square integrable, so $R_h \in \mathfrak{S}_2$. Thus all positive-time heat traces and their fixed polynomial energy moments below are finite: the decreasing inverse eigenvalues satisfy $r_j \le \|R_h\|_2/\sqrt{j}$, which already implies summability of $\exp[-tr_j^{-\gamma}]$ for every $t,\gamma>0$. The stronger estimate used for Bose singularities elsewhere is not an input to the argument in this appendix.

Set

$$\begin{aligned} v &= (\sqrt{\alpha_1},\dots,\sqrt{\alpha_M})^t,\\ P &= vv^*,\\ Q &= I_M - P,\\ z(s) &= \sqrt{w(s)}. \end{aligned}$$

The corresponding projections on $\mathcal{H}$ have the same names. The collapsed physical space is embedded as $L^2(\mathrm{d}s)\otimes v$. If R is the base-curve Green inverse conjugated from $L^2(w\mathrm{d}s)$ to $L^2(\mathrm{d}s)$, its aligned extension is $R_0 = R\otimes P$. The orthogonal Q space has inverse zero and heat zero. It represents an alignment complement, not physical modes of zero energy.

For $q>0$ define

$$\begin{aligned} T(q)_{ij} &= \sqrt{\alpha_i\alpha_j}\,\mathrm{e}^{-q|a_i-a_j|},\\ B(s,\xi) &= \frac{w(s)}{2|\xi|}T(|\xi|),\\ B_0(s,\xi) &= \frac{w(s)}{2|\xi|}P. \end{aligned} \tag{A3}$$

Fix $\gamma>0$ and $\tau>0$. With the value at zero defined to be zero, write

$$\begin{aligned} f_{\tau,\gamma}(b) &= \mathrm{e}^{-\tau b^{-\gamma}},\\ X_h &= R_h/h,\\ X_{0,h} &= R_0/h,\\ A_h(\tau) &= f_{\tau,\gamma}(X_h),\\ A_{0,h}(\tau) &= f_{\tau,\gamma}(X_{0,h}). \end{aligned}$$

These are the heat operators at inverse temperature $\beta=\tau h^\gamma$ for L_h^γ. In particular $\gamma=1/2$ weights single-excitation frequencies if L_h gives squared frequencies. Thus $H_h = L_h^\gamma$ is the one-particle Hamiltonian, and $A_h(\tau)=\exp(-\beta H_h)$ is its heat operator. Let $Z_h = \operatorname{Tr}A_h$, $Z_{0,h}=\operatorname{Tr}A_{0,h}$ and $\rho_h = A_h/Z_h$, $\rho_{0,h}=A_{0,h}/Z_{0,h}$. Put

$$F(s,\xi) = f_{\tau,\gamma}(B(s,\xi)), \tag{A4}$$

$$F_0(s,\xi) = \mathrm{e}^{-\tau(2|\xi|/w(s))^\gamma}P,$$

$$C_\gamma(\tau) = \frac{1}{2\pi}\int_\Gamma\int_{\mathbb{R}} \operatorname{tr}F(s,\xi)\,\mathrm{d}\xi\mathrm{d}s, \tag{A5}$$

$$G_\gamma = \Gamma(1+1/\gamma),$$

$$C_{\mathrm{col},\gamma}(\tau) = \frac{G_\gamma}{2\pi\tau^{1/\gamma}}. \tag{A6}$$

Lowercase tr denotes a finite matrix trace. Values at $\xi=0$ have no effect on these integrals.

The complete actual-operator conclusions are

$$hZ_h \longrightarrow C_\gamma(\tau), \tag{A7}$$

$$hZ_{0,h} \longrightarrow C_{\mathrm{col},\gamma}(\tau), \tag{A8}$$

$$\frac{1}{2}\|\rho_h-\rho_{0,h}\|_1 \longrightarrow \tag{A9}$$

$$D_\gamma(\tau) := \frac{1}{4\pi}\int_\Gamma\int_{\mathbb{R}} \tag{A10}$$

$$\operatorname{tr}\left|\frac{F}{C_\gamma(\tau)} - \frac{F_0}{C_{\mathrm{col},\gamma}(\tau)}\right|\mathrm{d}\xi\mathrm{d}s, \tag{A11}$$

$$\operatorname{Tr}(Q\rho_h) \longrightarrow p_{Q,\gamma}(\tau) := \frac{C_{Q,\gamma}(\tau)}{C_\gamma(\tau)}, \tag{A12}$$

$$C_{Q,\gamma}(\tau) = \frac{1}{2\pi}\int_\Gamma\int_{\mathbb{R}}\operatorname{tr}(QF)\,\mathrm{d}\xi\mathrm{d}s. \tag{A13}$$

For $M\ge 2$ these coefficients obey $0<p_{Q,\gamma}(\tau)\le D_\gamma(\tau)<1$. All data $M,a_i,\alpha_i,\Gamma,w,\Omega,\gamma,\tau$ are fixed before $h\downarrow 0$. Constants may depend on all of them. Endpoint limits in τ below are subsequent limits of these functions. No uniform claim is made when M grows, a weight or height separation vanishes, a mask approaches the tube, or τ is chosen as an arbitrary function of h.

1. A positive-resolvent comparison that retains every mode

We prove a slightly larger functional-calculus statement. Let g be real and continuous on $[0,\infty]$, with $g(0)=0$ and a finite limit of $g(x)/x^2$ at zero. For $\mathcal{T}(A)=A(I+A)^{-1}$ and $\Phi(y)=g(y/(1-y))$, the function $\Phi(y)/y^2$ is continuous on $[0,1]$. For every $\eta>0$ a polynomial p therefore satisfies

$$|\Phi(y)-y^2p(y)|\le\eta y^2 \quad (0\le y\le 1). \tag{A14}$$

Suppose $U=\mathcal{T}(A)$, $V=\mathcal{T}(D)$ for positive compact operators, with $\operatorname{Tr}U^2+\operatorname{Tr}V^2\le C/h$, and $U-V=J+E$, where $\operatorname{rank}J\le r$, $\|J\|\le C_J$ and $\|E\|_2\le e$. For a monomial of degree $k\ge 2$ use

$$U^k-V^k=\sum_{j=0}^{k-1}U^j(J+E)V^{k-1-j}.$$

Each J term has trace norm at most C_Jr. In each E term at least one outside exponent is positive, and the corresponding factor has Hilbert–Schmidt norm at most $\max(\|U\|_2,\|V\|_2)$. Schatten Holder's inequality then gives

$$h\,\|g(A)-g(D)\|_1 \le C\eta + C_p\big(hC_Jr+\sqrt{h}\,e\big). \tag{A15}$$

Here the two approximation errors in (A14) cost at most $\eta\operatorname{Tr}(U^2+V^2)$, by spectral calculus.

For a decomposition $A-D=J_{\mathrm{raw}}+E_{\mathrm{raw}}$, the exact identity

$$\mathcal{T}(A)-\mathcal{T}(D)=(I+A)^{-1}(A-D)(I+D)^{-1} \tag{A16}$$

preserves the rank bound and does not increase the residual Hilbert–Schmidt norm. If $\|E_{\mathrm{raw}}\| \le C$ uniformly, the transformed finite-rank term has norm at most $1 + C$, since the difference of two positive contractions has norm at most one. Even without that operator-norm bound, a uniformly bounded rank-r decomposition is available. Set $K = U - V$ and keep its r largest singular values, with their signs, to obtain J. Then $\|J\| \le 1$. The squared Hilbert–Schmidt residual is the sum of the remaining squared singular values, and is no larger than for any other rank-r approximation; applying this to the rank-r term already supplied by the resolvent identity gives $\|K - J\|_2 \le \|E_{\mathrm{raw}}\|_2$. For completeness, that minimizing property follows by projecting onto the range of any proposed rank-r approximation: its discarded Hilbert–Schmidt norm is at least the sum after the largest r eigenvalues of KK^*, by the variational principle for their partial sum. Spectral truncation attains the bound. The difference K is Hilbert–Schmidt already, because the resolvent identity writes it as a finite-rank term plus such a residual. Thus one may always take $C_J = 1$ after this additional truncation; no bound on the size of J_{raw} is needed. If only the budget for V is initially known,

$$\|U\|_2 \le \|V\|_2 + C_J\sqrt{r} + e$$

supplies it whenever the displayed right side is $O(h^{-1/2})$. In particular $hr = o(1)$ and $\sqrt{h}\,e = o(1)$ imply a whole-operator $o(h^{-1})$ trace-norm comparison. This proof uses no scalar Lipschitz assertion in trace norm and no short-time extrapolation of a fixed-temperature bound.

2. The curved kernel and a manifestly positive model

Write $k_n = 2\pi n/\ell = \varrho n$. Define the positive matrix convolution operator $\mathcal{C}_h$ by its Fourier multipliers

$$\begin{aligned} \mathcal{C}_h(0) &= I_M, \\ \mathcal{C}_h(n) &= \frac{T(h|k_n|)}{2|k_n|} \\ &\quad (n \ne 0), \\ \mathcal{A}_h &= h^{-1} M_z \mathcal{C}_h M_z. \end{aligned} \tag{A17}$$

For the collapsed model use the scalar convolution $\mathcal{C}_0$ with multipliers $\mathcal{C}_0(0) = 1$, $\mathcal{C}_0(n) = (2|k_n|)^{-1}$ for $n \ne 0$, and $\mathcal{A}_{0,h} = h^{-1}(M_z \mathcal{C}_0 M_z) \otimes P$. The finitely many zero-frequency choices affect no leading trace. Positivity follows from

$$\mathrm{e}^{-q|a-b|} = \frac{1}{\pi}\int_{\mathbb{R}} \frac{q\mathrm{e}^{it(a-b)}}{q^2 + t^2}\,\mathrm{d}t.$$

A nonzero finite exponential sum at distinct heights is not identically zero, so the associated quadratic form is strictly positive for $q > 0$. Consequently $T(q) > 0$, $\operatorname{tr} T(q) = 1$, and $\|T(q)\| \le 1$. At most M eigenvalues occur at each Fourier index and each is $O((1 + |n|)^{-1})$. Min–max after multiplication by bounded z yields

$$\begin{aligned} &\operatorname{Tr}\mathcal{T}(\mathcal{A}_h)^2 + \operatorname{Tr}\mathcal{T}(\mathcal{A}_{0,h})^2 \\ &\le C\sum_{j\ge 1} \min\{1, (hj)^{-2}\} \le C'/h. \end{aligned} \tag{A18}$$

The regular part $G_\Omega(x,y) + (2\pi)^{-1}\log|x-y|$ is smooth on the product of the closed tube with itself. Both partial Laplacians vanish distributionally there; their sum is elliptic in four variables, so interior regularity includes the diagonal. This argument is local and does not require smooth distant boundary components. Define

$$\begin{aligned} \ell_*(u) &= \frac{\ell}{\pi}\left|\sin\frac{\pi u}{\ell}\right|, \\ \Lambda_\delta(u) &= -\frac{1}{4\pi}\log\left(1 + \frac{\delta^2}{\ell_*(u)^2}\right), \\ D(s,t) &= n(s)\cdot\nabla_x G_\Omega(\mathbf{r}(s), \mathbf{r}(t)). \end{aligned}$$

The latter off-diagonal expression extends smoothly: the normal numerator in the logarithmic part vanishes to order $(s-t)^2$, as does the chord-length squared denominator. The ratio of the geometric chord to $\ell_*(s-t)$ is smooth and positive across the diagonal. It follows that $R = M_z\mathcal{C}_0 M_z + \mathcal{S}_0$, with $\mathcal{S}_0$ a smooth-kernel operator.

For $\delta = h|a_i - a_j|$, the required uniform kernel expansion is

$$G_\Omega(T_{i,h}(s), T_{j,h}(t)) \tag{A19}$$

$$= G_\Omega(\mathbf{r}(s), \mathbf{r}(t)) + \Lambda_\delta(s-t) \tag{A20}$$

$$+h\{a_i D(s,t) + a_j D(t,s)\} + \mathcal{E}_{h,ij}(s,t),$$

$$|\mathcal{E}_{h,ij}(s,t)| \le Ch^2 + Ch\frac{\delta^2}{u^2 + \delta^2}, \tag{A21}$$

where u is the short periodic coordinate, and the fraction is zero when $\delta = 0$. Here and below kernels may be specified off their null diagonal. To verify the singular part, put $p = \mathbf{r}(s) - \mathbf{r}(t)$, $b = n(s) - n(t)$, $c = n(s) + n(t)$, $m = (a_i + a_j)/2$, and $d = (a_i - a_j)/2$. The changed chord is $p + h(mb + dc)$ and $b \cdot c = 0$. The smooth quotients satisfy

$$\begin{aligned} \frac{p\cdot b}{|p|^2} &= O(1), \\ \frac{p\cdot c}{|p|^2} &= O(|u|), \\ \frac{|b|^2}{|p|^2} &= O(1), \\ \frac{|c|^2}{|p|^2} &= \frac{4}{\ell_*(u)^2} + O(1). \end{aligned}$$

These follow by Taylor expansion of an arclength parametrization and $n \perp \mathbf{r}'$. Expand the logarithm of $|p + h(mb + dc)|^2/|p|^2$ about $1 + \delta^2/\ell_*(u)^2$. The relative perturbation is $O(h)$, its quadratic logarithm error is $O(h^2)$, and replacing the denominator of the linear term

by one costs at most the second term in (A21). The smooth regular part and points away from the diagonal have the ordinary $O(h^2)$ Taylor remainder. This proves the estimate, including unequal, uncentered profiles. A uniformly $O(h^2)$ pointwise remainder would in general be false. Squaring and integrating its strip term gives $O(h^2\delta) = O(h^3)$; hence the weighted matrix remainder has Hilbert–Schmidt norm $O(h^{3/2})$.

The exact Fourier multipliers of Λ_δ, for convolution against $\mathrm{d}s$, are

$$\begin{aligned} b_\delta &= 2\,\mathrm{arsinh}(\varrho\delta/2), \\ \Lambda_\delta(0) &= -\frac{b_\delta}{2\varrho}, \\ \Lambda_\delta(n) &= -\frac{1-\mathrm{e}^{-b_\delta|n|}}{2\varrho|n|} \\ &\quad (n \neq 0). \end{aligned} \tag{A22}$$

For example factor $\sin^2(\theta/2) + \sinh^2(b_\delta/2) = (\cosh b_\delta - \cos\theta)/2$ and expand the logarithm in its integrable Fourier series. Its constant term is b_δ, and its cosine coefficient is $2(1-\mathrm{e}^{-b_\delta n})/n$, giving precisely (A22). For each fixed nonzero height difference,

$$\begin{aligned} |b_{h|a_i-a_j|} - \varrho h|a_i - a_j|| &\le Ch^3, \\ \min\{b_{h|a_i-a_j|}, \varrho h|a_i-a_j|\} &\ge ch. \end{aligned}$$

The mean value theorem therefore bounds the discrepancy of nonzero Fourier multipliers by $Ch^3\mathrm{e}^{-ch|n|}$. Its squared sum is $O(h^5)$; diagonal matrix entries have zero discrepancy. One must use the positive model (A17), not assume positivity of the matrix obtained from the exact periodic replacement. Collecting terms gives

$$\begin{aligned} R_h = M_z\mathcal{C}_h M_z + \mathcal{S}_0 \otimes P + h\mathcal{S}_1 + J_{0,h} + \mathcal{R}_h + \mathcal{E}_h, \\ \operatorname{rank} J_{0,h} \le M, \\ \|\mathcal{R}_h\|_2 = O(h^{5/2}), \\ \|\mathcal{E}_h\|_2 = O(h^{3/2}). \end{aligned} \tag{A23}$$

Here $\mathcal{S}_1$ is the fixed smooth matrix kernel formed from the normal derivatives. The zero-frequency discrepancy, sandwiched by z, has range in $\mathrm{span}\{z\} \otimes \mathbb{C}^M$, giving the stated rank. Its norm will not be needed.

If Π_N projects onto $|n| \le N$ in every component, any fixed smooth matrix kernel $\mathcal{S}$ satisfies $\|\mathcal{S} - \Pi_N\mathcal{S}\Pi_N\|_2 \le C_k N^{-k}$ for every integer k, by repeated integration by parts in both Fourier indices. Take $N = \lceil h^{-1/2}\rceil$, $k \ge 4$, and divide (A23) by h. The result is a finite-rank term of rank $O(h^{-1/2})$ and a Hilbert–Schmidt residual $O(h^{1/2})$. Equations (A16) and (A18) first give the actual resolvent-square budget and then (A15) gives

$$\begin{aligned} h\,\|g(X_h) - g(\mathcal{A}_h)\|_1 &\longrightarrow 0, \\ h\,\|g(X_{0,h}) - g(\mathcal{A}_{0,h})\|_1 &\longrightarrow 0. \end{aligned} \tag{A24}$$

Both operators compared are positive. No positivity is attributed to an intermediate truncated geometric perturbation.

3. Matrix Fourier words, low and high frequencies, and sharp bands

Choose $0 < 2d < R$ and a smooth $0 \le \chi \le 1$ supported where $d \le |\xi| \le 2R$, equal to one where $2d \le |\xi| \le R$. Replace the nonzero multipliers of $\mathcal{C}_h/h$ by $\chi(hk_n)T(|hk_n|)/(2|hk_n|)$, and set the zero multiplier to zero. Keep the two multiplications by z; call the result $\mathcal{A}_h^{d,R}$. Use the same construction for the reference. These positive operators have rank $O(R/h)$ and uniformly bounded norm at fixed d, R. They are below their full models in operator order, so eigenvalue comparison preserves the resolvent-square trace budget. The discarded low-frequency part has rank $O(d/h) + O(1)$. The high-frequency part obeys

$$\|E_{\mathrm{hi}}\| \le C/R, \qquad \|E_{\mathrm{hi}}\|_2^2 \le C/(hR). \tag{A25}$$

Indeed these follow by summing $(h|n|)^{-2}$ over $|hk_n| > R$ and using $\|T(q)\| \le 1$; bounded multiplication by z changes only constants. After the resolvent identity the finite-rank part has norm at most $1+C/R$, independently of d. Consequently

$$\limsup_{h\downarrow 0} h\left\|g(\mathcal{A}_h) - g(\mathcal{A}_h^{d,R})\right\|_1 \le C\eta + C_p(d + R^{-1/2}), \tag{A26}$$

and identically for the reference. For each fixed approximation polynomial first let $d \downarrow 0$, $R \uparrow \infty$, and then $\eta \downarrow 0$. This controls all original Fourier modes outside the annulus; no arbitrary eigenvalue reindexing is involved.

Here is the trace calculation on the fixed annulus. Expand a word in operators $M_{z_j} m_j(hD_s) M_{z_j}$, with smooth scalar z_j and smooth compactly supported matrix multipliers m_j, in the basis $\ell^{-1/2}\mathrm{e}^{ik_n s}$. Each multiplication shifts the index by an integer and has rapidly decreasing, absolutely summable Fourier coefficients. For any fixed list of shifts r_j, $m_j(hk_{n+r_j}) - m_j(hk_n) = o(1)$ uniformly in n. The number of contributing n is $O(h^{-1})$ uniformly in the shifts, by anchoring to any one compactly supported multiplier factor; the remaining factors are bounded. The absolute sum of the products of Fourier coefficients is finite. Dominated convergence therefore permits summing over the shift lists. The zero-total-shift condition reconstructs the spatial average of the position factors. The remaining frequency sum is a Riemann sum of spacing $2\pi h/\ell$. Thus, retaining matrix order,

$$\begin{aligned} &h\,\mathrm{Tr}(\mathcal{B}_{1,h}\cdots\mathcal{B}_{r,h}) \\ &\longrightarrow \frac{1}{2\pi}\int_\Gamma\int_{\mathbb{R}} \mathrm{tr}(b_1(s,\xi)\cdots b_r(s,\xi))\,\mathrm{d}\xi\mathrm{d}s. \end{aligned} \tag{A27}$$

The same calculation permits smooth scalar position insertions, fixed matrix insertions, and a smooth bounded Fourier multiplier $\psi(hD_s)$: its shifted values also converge uniformly for any fixed shift on the relevant compact frequency set. No factors have been commuted in this calculation.

Uniform polynomial approximation on the bounded spectra of the cutoff models, with approximating polynomials zero at zero, extends (A27) to words in their g functions. The error cost after multiplication by h is $O(R\eta)$, since all ranges lie in a common space of dimension $O(R/h)$. The difference of two such self-adjoint functions need not commute with either factor. Approximate the continuous function $|x|$ on its bounded spectral interval by polynomials zero at zero, using that same finite range. This proves, for any real c, c_0,

$$h \operatorname{Tr}\left|c\, g(\mathcal{A}_h^{d,R}) - c_0 g(\mathcal{A}_{0,h}^{d,R})\right| \longrightarrow \frac{1}{2\pi}\int \operatorname{tr}\left|c\, g(\chi B) - c_0 g(\chi B_0)\right| \mathrm{d}\xi \mathrm{d}s.$$

On removing cutoffs, the low-frequency symbol cost is $O(d)$ because g is bounded. Since $g(b) = O(b^2)$ near zero and $\|B\| \le w_{\max}/(2|\xi|)$, the high-frequency symbol has an integrable $C|\xi|^{-2}$ bound, uniformly in χ. Use (A26) on the operator side and $|\operatorname{Tr}|X| - \operatorname{Tr}|Y|| \le \|X - Y\|_1$ for absolute traces. Finally (A24) transfers all conclusions to the actual geometry. In particular

$$h \operatorname{Tr}[(M_q \otimes V) g(X_h)] \tag{A28}$$

$$\longrightarrow \frac{1}{2\pi}\int q(s) \operatorname{tr}[V g(B)] \mathrm{d}\xi \mathrm{d}s, \tag{A29}$$

$$h \operatorname{Tr}\left|c\, g(X_h) - c_0 g(X_{0,h})\right| \tag{A30}$$

$$\longrightarrow \frac{1}{2\pi}\int \operatorname{tr}\left|c\, g(B) - c_0 g(B_0)\right| \mathrm{d}\xi \mathrm{d}s, \tag{A31}$$

for smooth q and fixed matrix V, with the analogous reference law. Fourier insertions described above are also allowed. All estimates include the singular low-frequency end and the infinite high-frequency tail. For heat a stronger symbol bound is

$$0 \le F(s,\xi) \le \mathrm{e}^{-\tau(2|\xi|/w_{\max})^{\gamma}} I_M. \tag{A32}$$

This is spectral calculus against a scalar multiple of I_M, not operator monotonicity of the exponential.

For later receiver constructions it is useful to include sharp Fourier bands explicitly. Let $\Pi_{h;d,R}$ project onto $d < |hk_n| < R$ in every component, where $0 \le d < R < \infty$. For either heat operator and any fixed matrix V,

$$h \operatorname{Tr}[\Pi_{h;d,R} V A_h(\tau)] \longrightarrow \frac{1}{2\pi}\int_\Gamma \int_{d<|\xi|<R} \operatorname{tr}(VF)\, \mathrm{d}\xi \mathrm{d}s. \tag{A33}$$

To see this, approximate the band indicator by a smooth Fourier multiplier differing only in intervals of total length $O(\delta)$ around its endpoints. The difference has norm at most one and rank $O(\delta/h) + O(1)$. Because $\|A_h\| \le 1$, its trace against VA_h costs at most $\|V\|$ times this rank; after multiplying by h this is $O(\delta) + O(h)$. The symbol edge cost has the same bound. Let $h \downarrow 0$ and then $\delta \downarrow 0$. The possible $n = 0$ ambiguity costs only M dimensions. The smooth insertion argument applies to the full heat by (A26) and (A24). Subtracting the $|hk_n| \le R$ law from the complete trace law proves the $|hk_n| > R$ law as well. In particular its $V = Q$ or $V = I_M$ heat coefficient tends to zero as $R \uparrow \infty$ by (A32).

Taking $g = f_{\tau,\gamma}$ now proves (A8) and the numerator in (A13); the reference integral is elementary and uses $\int w \mathrm{d}s = 1$. If $c_h = hZ_h$ and $c_{0,h} = hZ_{0,h}$, then $\rho_h - \rho_{0,h} = h(A_h/c_h - A_{0,h}/c_{0,h})$. Replacing these denominators by their positive limits costs $o(1)$ in trace norm, since each heat operator is positive. Apply (A31) with the limiting denominators to obtain (A11). For $M \ge 2$, F is positive definite at every $\xi \ne 0$, so $\operatorname{tr}(QF) > 0$. The variational characterization of trace distance with the effect Q gives $p_Q \le D$, because $Q\rho_{0,h} = 0$. Finally positive matrices X, Y satisfy $\operatorname{tr}|X - Y| \le \operatorname{tr} X + \operatorname{tr} Y$, with equality exactly when their supports are orthogonal. This follows by evaluating $X - Y$ on its positive and negative spectral projections: equality forces each positive matrix into the corresponding disjoint subspace. Here F/C_γ is positive definite and F_0/C_{col} is nonzero, so the inequality is strict pointwise. The integral of their summed traces is two, proving $D < 1$.

Appendix B: Growing-index counting, a strict deficit, and transverse rotation

For fixed s, order the eigenvalues of $B(s,q)$ as $b_1 > \cdots > b_M > 0$ and put $e_j = 1/b_j$, with simplicity justified below. The free one-dimensional resolvent gives

$$x^* \frac{T(q)}{2q} x = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{|\sum_j \sqrt{\alpha_j} x_j \mathrm{e}^{ita_j}|^2}{t^2 + q^2}\, \mathrm{d}t. \tag{B1}$$

Subtracting this formula at $q_2 > q_1 > 0$ proves a positive definite difference. Thus all ordered $b_j(s,q)$ strictly decrease and all $e_j(s,q)$ strictly increase. This is a statement about $T(q)/(2q)$; the eigenvalues of $T(q)$ itself cannot all decrease since its trace is one.

Set $\Delta_i = a_{i+1} - a_i > 0$ and $A_\alpha = \operatorname{diag} \alpha_i$. The exact energy matrix is

$$B(s,q)^{-1} = w(s)^{-1} A_\alpha^{-1/2} J(q) A_\alpha^{-1/2}, \tag{B2}$$

where, for $M \ge 2$,

$$J_{11} = q[1 + \coth(q\Delta_1)],$$
$$J_{MM} = q[1 + \coth(q\Delta_{M-1})],$$
$$J_{ii} = q[\coth(q\Delta_{i-1}) + \coth(q\Delta_i)] \quad (1 < i < M),$$
$$J_{i,i+1} = J_{i+1,i} = -q \operatorname{csch}(q\Delta_i),$$

and the other entries vanish. For $M = 1$ use $J(q) = 2q$. An energy derivation fixes every constant. Minimize $\int_{\mathbb{R}} (|u'|^2 + q^2|u|^2) \mathrm{d}a$ subject to $u(a_i) = x_i$. On the interval of length Δ_i, the hyperbolic interpolation solving $-u'' + q^2 u = 0$ has energy

$$q \coth(q\Delta_i)(|x_i|^2 + |x_{i+1}|^2) - 2q \operatorname{csch}(q\Delta_i) \operatorname{Re}(\overline{x_i} x_{i+1}).$$

The two exterior half-lines contribute $q|x_1|^2 + q|x_M|^2$. The reproducing kernel for the same energy is $\mathrm{e}^{-q|a-b|}/(2q)$, so its evaluation Gram matrix has inverse $J(q)$. Conjugating by the masses proves (B2).

As $q \downarrow 0$, $J(q)$ tends to the nearest-neighbor path Laplacian $J(0)$ with quadratic form $\sum_i |x_{i+1} - x_i|^2/\Delta_i$. Write

$$\Lambda_\alpha = A_\alpha^{-1/2} J(0) A_\alpha^{-1/2}, \qquad 0 = \ell_1 < \ell_2 < \cdots < \ell_M. \tag{B3}$$

The kernel is spanned by v. Positivity of the other eigenvalues follows from the quadratic form, and simplicity follows from the irreducible symmetric tridiagonal recurrence: an eigenvector is determined by its first component, which cannot vanish. The same argument applies for $q > 0$ and proves simplicity in the definition of e_j. Consequently

$$\begin{gathered} e_j(s, 0+) = \ell_j / w(s), \\ e_j(s, q) \geq 2q / w(s), \\ e_j(s, q) \longrightarrow \infty \\ (q \to \infty). \end{gathered}$$

The bound uses $b_j \leq \operatorname{tr} B = w/(2q)$. Every branch therefore has exactly one positive-width energy cutoff at energy E when $w(s)E > \ell_j$, and none when $w(s)E \leq \ell_j$.

For $g > 0$ let $-\kappa_j(g)^2$ be the negative eigenvalues of the attractive delta Hamiltonian

$$-\frac{\mathrm{d}^2}{\mathrm{d}a^2} - g \sum_i \alpha_i \delta_{a_i} \quad \text{on } L^2(\mathbb{R}),$$

defined by the form $\int |u'|^2 \mathrm{d}a - g \sum_i \alpha_i |u(a_i)|^2$ on $H^1(\mathbb{R})$. Point evaluation is infinitesimally form bounded: for instance $|u(a)|^2 \leq \|u\|_2 \, \|u'\|_2$, followed by Young's inequality, gives an arbitrarily small coefficient of $\|u'\|_2^2$. Thus the form is closed and lower bounded. Its negative index is at most M, because the perturbation factors through M evaluations. Factoring those evaluations through $(-\mathrm{d}^2/\mathrm{d}a^2 + q^2)^{-1/2}$ shows that the negative index after adding q^2 equals the number of eigenvalues of $gT(q)/(2q)$ above one; the nonzero spectra of XX^* and X^*X coincide. Equality with one also gives the eigenspace at energy $-q^2$ by the weak equation. The top eigenvalue diverges as $q \downarrow 0$, so a lowest negative state always exists. We set $\kappa_j(g) = 0$ for absent branches. The preceding strictly monotone energy branches imply

$$\#\{j : b_j(s, q) \geq E^{-1}\} = \#\{j : \kappa_j(w(s)E) \geq q\} \quad (q > 0), \tag{B4}$$

including the equality threshold. In particular the number of negative states is $\#\{j : \ell_j < g\}$; equality at a threshold does not add a negative state.

The actual growing-index count is

$$\begin{gathered} h\#\{k : \lambda_k(L_h) \leq E/h\} \longrightarrow \\ \mathcal{N}_\alpha(E) = \frac{1}{\pi} \int_\Gamma \sum_{j=1}^{M} \kappa_j(w(s)E) \, \mathrm{d}s, \\ E > 0. \end{gathered} \tag{B5}$$

To pass rigorously from continuous tests to this cutoff, approximate $1_{[1/E,\infty]}$ on $[0, \infty]$ from above and below by continuous functions zero near zero. They belong to the class used in (A29). For each s, strict monotonicity gives at most one positive q with $b_j(s, q) = 1/E$. Fubini proves that the phase-space level has measure zero. The transition interval is supported in a bounded frequency region, because $b_j \leq w_{\max}/(2q)$. Dominated convergence squeezes the two phase-space approximations together. Integrating (B4) then gives (B5). This proof controls the whole growing count, rather than a sum of fixed-eigenvalue approximations. For L_h^γ below U/h^γ, replace E by $U^{1/\gamma}$.

1. A self-contained finite-profile form of the classical sharp bound

The inequality $\sum_j \kappa_j(g) \leq g/2$, with equality only for a single point mass, is the sharp measure Lieb–Thirring result of Hundertmark, Lieb, and Thomas, Theorem 5 of Ref. [3]. Here is a finite-matrix derivation sufficient for the present profile, so the counting application does not require a measure approximation.

Write $\mathcal{L}(q) = gT(q)$ and $\Sigma_j(q) = \sum_{i=1}^{j} \lambda_i(\mathcal{L}(q))$. For $q_2 > q_1 \geq 0$, set $t = q_2 - q_1$ and $U_\zeta = \operatorname{diag}(\mathrm{e}^{i\zeta a_i})$. The Fourier transform of the Cauchy density gives the exact unitary average

$$\mathcal{L}(q_2) = \int_{\mathbb{R}} U_\zeta \mathcal{L}(q_1) U_\zeta^* \frac{t \, \mathrm{d}\zeta}{\pi(t^2 + \zeta^2)}.$$

The sum of the j largest eigenvalues of a Hermitian matrix is $\sup\{\operatorname{tr}(EX) : 0 \leq E \leq I, \operatorname{tr} E = j\}$, which is convex and unitarily invariant. Therefore $\Sigma_j(q_2) \leq \Sigma_j(q_1)$. For $M \geq 2$ the top eigenvalue is strictly decreasing: the positive matrix $\mathcal{L}(q_2)$ has a strictly positive top eigenvector, and every off-diagonal entry strictly increases on changing q_2 to q_1. Its Rayleigh quotient strictly increases as well.

Order $\kappa_1 > \kappa_2 > \cdots$ over the existing states and append zeros to length M. The energy simplicity established above gives the strict first inequality. The Birman–Schwinger equation gives $2\kappa_j = \lambda_j(\mathcal{L}(\kappa_j))$ for existing states. For absent states this is also true: $\mathcal{L}(0) = gP$ has all eigenvalues after the first equal to zero. Telescoping partial sums gives the exact identity

$$\begin{aligned} &2 \sum_{j=1}^{M} \kappa_j \\ &= \Sigma_M(\kappa_M) + \sum_{j=1}^{M-1} [\Sigma_j(\kappa_j) - \Sigma_j(\kappa_{j+1})] \\ &\leq g + \Sigma_1(\kappa_1) - \Sigma_1(\kappa_2) < g. \end{aligned} \tag{B6}$$

The trace $\Sigma_M(q)$ equals g at every q. Every bracket is nonpositive, and the first is strictly negative, including when only one negative state exists and $\kappa_2 = 0$. For

$M = 1$ directly $\kappa_1 = g/2$. Repeated heights, if initially allowed, are first merged; equality is therefore exactly the single-point case. The sharp inequality and its rigidity are classical; their use here is after the actual physical-symbol identification.

Define $\delta_\alpha(g) = g/2 - \sum_j \kappa_j(g)$. Equations (B5) and (B6) imply

$$\begin{gathered} 0 < \mathcal{N}_\alpha(E) < \frac{E}{2\pi}, \\ \frac{E}{2\pi} - \mathcal{N}_\alpha(E) = \frac{1}{\pi}\int_\Gamma \delta_\alpha(w(s)E)\,\mathrm{d}s \\ (M \geq 2). \end{gathered} \tag{B7}$$

The symbol heat is the Stieltjes integral against this counting function. Integration by parts, or positive layer-cake integration, gives

$$C_\gamma(\tau) = \tau\gamma \int_0^\infty E^{\gamma-1}\mathrm{e}^{-\tau E^\gamma}\mathcal{N}_\alpha(E)\,\mathrm{d}E,$$

$$C_{\mathrm{col},\gamma}(\tau) - C_\gamma(\tau) \tag{B8}$$

$$= \frac{\tau\gamma}{\pi}\int_\Gamma\int_0^\infty E^{\gamma-1}\mathrm{e}^{-\tau E^\gamma}\delta_\alpha(w(s)E)\,\mathrm{d}E\mathrm{d}s > 0. \tag{B9}$$

There are no boundary terms because $0 \leq \mathcal{N}_\alpha(E) \leq E/(2\pi)$ and $\mathcal{N}_\alpha(0) = 0$. This proves strict partition deficit for every finite $\tau > 0$ and every $\gamma > 0$, with equality only for the merged single-point profile.

2. Algebraic transverse occupation from the lowest branch

Define profile and density statistics

$$\begin{aligned} d_i &= \sum_j \alpha_j |a_i - a_j|, \\ G &= \sum_i \alpha_i d_i, \\ V &= \sum_i \alpha_i (d_i - G)^2, \\ I_k &= \int_\Gamma w(s)^k\,\mathrm{d}s. \end{aligned} \tag{B10}$$

Let $\mathsf{D}_{ij} = \sqrt{\alpha_i\alpha_j}|a_i - a_j|$. The finite matrix $T(q) = P - q\mathsf{D} + O(q^2)$ is analytic at zero. Its simple top eigenpair, normalized by $v^* u(q) > 0$, therefore obeys

$$\begin{aligned} \mu(q) &= 1 - Gq + O(q^2), \\ u(q) &= v - qQ\mathsf{D}v + O(q^2), \\ \|Qu(q)\|^2 &= Vq^2 + O(q^3). \end{aligned} \tag{B11}$$

These coefficients also follow directly by differentiating $T(q)u(q) = \mu(q)u(q)$: the v component gives $\mu'(0) = -G$; the Q component and normalization give $u'(0) = -Q\mathsf{D}v$; and $(Q\mathsf{D}v)_i = \sqrt{\alpha_i}(d_i - G)$. For small g only the first bound state exists, by (B3). Its equation $2\kappa = g\mu(\kappa)$ and the implicit function theorem give

$$\kappa(g) = \frac{g}{2} - \frac{G}{4}g^2 + O(g^3), \qquad \kappa'(g) = \frac{1}{2} - \frac{G}{2}g + O(g^2). \tag{B12}$$

For $M = 1$ the same expressions hold with $G = V = 0$. The mean-distance coefficient G is the familiar one-dimensional weak-coupling statistic [17, 18]; see Theorem 8.5 and Eq. (9.24) of Ref. [18]. The finite atomic coefficient used here has been derived directly; its occurrence alone is not a new scalar invariant.

Choose a small fixed energy E_0 below $\ell_2/w_{\max}$ when $M \geq 2$, and small enough for (B12) uniformly over s. On the lowest branch put $q = \kappa(w(s)E)$. Then

$$\begin{aligned} \frac{\mathrm{d}q}{\mathrm{d}E} &= \frac{w(s)}{2} - \frac{Gw(s)^2}{2}E + O(E^2), \\ \|Qu(\kappa(w(s)E))\|^2\,\frac{\mathrm{d}q}{\mathrm{d}E} &= \frac{Vw(s)^3}{8}E^2 + O(E^3), \end{aligned} \tag{B13}$$

with uniform errors. The Taylor formulas are used only on that fixed small interval. To bound the rest of every branch, put $a_* = \ell_2/w_{\max} > 0$. For $j \geq 2$, $e_j(s,q) \geq a_*$ and $e_j(s,q) \geq 2q/w_{\max}$, whence

$$\begin{aligned} C_{\mathrm{exc},\gamma}(\tau) &:= \frac{1}{\pi}\int_\Gamma\int_0^\infty \sum_{j=2}^{M} \mathrm{e}^{-\tau e_j(s,q)^\gamma}\,\mathrm{d}q\mathrm{d}s \\ &\leq C\tau^{-1/\gamma}\mathrm{e}^{-\tau a_*^\gamma/2}. \end{aligned} \tag{B14}$$

This follows from the integrable product bound $\mathrm{e}^{-\tau e_j^\gamma} \leq \mathrm{e}^{-\tau a_*^\gamma/2}\mathrm{e}^{-(\tau/2)(2q/w_{\max})^\gamma}$. The portion of the lowest branch with $E \geq E_0$ has the identical bound with a_* replaced by E_0, by integrating in q. Thus the infinite tails are exponentially small, including their phase volume.

Now use $\int_0^\infty E^r\mathrm{e}^{-\tau E^\gamma}\mathrm{d}E = \Gamma((r+1)/\gamma)\tau^{-(r+1)/\gamma}/\gamma$ for $r > -1$. Integrating (B13) and its uniformly bounded remainders gives

$$C_\gamma(\tau) = \frac{G_\gamma}{2\pi}\tau^{-1/\gamma} - \frac{GI_2\Gamma(1+2/\gamma)}{4\pi}\tau^{-2/\gamma} + O(\tau^{-3/\gamma}), \tag{B15}$$

$$\begin{aligned} C_{Q,\gamma}(\tau) &= \frac{VI_3\Gamma(1+3/\gamma)}{24\pi}\tau^{-3/\gamma} + O(\tau^{-4/\gamma}), \\ p_{Q,\gamma}(\tau) &= \frac{VI_3\Gamma(1+3/\gamma)}{12G_\gamma}\tau^{-2/\gamma} + O(\tau^{-3/\gamma}). \end{aligned} \tag{B16}$$

For $\gamma = 1$ the latter leading coefficient is $VI_3/2$; for $\gamma = 1/2$ it is $30VI_3$. The excited fraction in (B14), divided by C_γ, is exponentially small. The algebraic term in (B16) comes instead from the rotation of the lowest symbol eigenspace relative to the fixed comparison P. These are limiting symbol branches, not a claim of exact global band labels for individual curved eigenstates.

The vanishing cases of V can be determined exactly. With $F_i = \sum_{j\leq i}\alpha_j$,

$$d_{i+1} - d_i = \Delta_i(2F_i - 1).$$

Since all weights and gaps are positive, $V = 0$ means all d_i coincide. For $M \geq 2$ this forces every F_i to equal $1/2$, which is possible only for $M = 2$ with equal weights. For that profile $T(q)$ has constant row sums and v is its top eigenvector at every q; the lowest branch has exactly zero Q weight and $C_Q = C_{\mathrm{exc}}$ is exponentially small. For $M = 1$, $Q = 0$. Every unequal two-layer profile and every profile with $M \geq 3$ has $V > 0$ and the algebraic leading term is strictly positive.

Appendix C: One-excitation scalar agreement and recoverable information

For each fixed integer $p \geq 0$ the test $g_{p,\tau,\gamma}(b) = (\tau b^{-\gamma})^p \mathrm{e}^{-\tau b^{-\gamma}}$ with value zero at $b = 0$ belongs to the class already proved. It is exponentially flat at zero and has a finite limit at infinity, equal to one for $p = 0$ and zero for $p > 0$. Thus, independently of differentiating any asymptotic remainder,

$$\begin{aligned} h\,\mathrm{Tr}[(\tau X_h^{-\gamma})^p A_h(\tau)] \\ \longrightarrow C_{p,\gamma}(\tau) := \\ \frac{1}{2\pi}\int_\Gamma \int_{\mathbb{R}} \\ \mathrm{tr}[(\tau B^{-\gamma})^p \mathrm{e}^{-\tau B^{-\gamma}}]\,\mathrm{d}\xi \mathrm{d}s, \\ C_{0,\gamma} = C_\gamma. \end{aligned} \tag{C1}$$

Products with inverse powers here denote their bounded spectral functions, zero on the inverse kernel. The same result holds for each fixed real $p \geq 0$ if desired. The high-frequency estimate is $x^p \mathrm{e}^{-x} \leq C_p \mathrm{e}^{-x/2}$; the low-frequency estimate is simply boundedness of this scalar test. These also justify differentiation of the limiting symbol integrals: $C_{1,\gamma} = -\tau C'_\gamma$ and $C_{2,\gamma} = \tau^2 C''_\gamma$.

For every fixed $r > 0$, $r \neq 1$, the exact Gibbs identity $\mathrm{Tr}\,\rho_h^r = Z_h(r\tau)/Z_h(\tau)^r$ gives the Renyi entropy

$$S_r(\rho_h) = \log(1/h) + \frac{\log C_\gamma(r\tau) - r \log C_\gamma(\tau)}{1-r} + o(1). \tag{C2}$$

The von Neumann entropy is treated separately, using its exact Gibbs identity and the $p = 1$ case of (C1):

$$S(\rho_h) = \log(1/h) + \log C_\gamma(\tau) + \frac{C_{1,\gamma}(\tau)}{C_\gamma(\tau)} + o(1). \tag{C3}$$

Their finiteness at fixed h follows from the spectral exponential estimate given after (A2); no dimension-free entropy continuity is used. Likewise

$$\beta\langle L_h^\gamma\rangle \longrightarrow \frac{C_{1,\gamma}}{C_\gamma}, \qquad \beta^2\,\mathrm{Var}(L_h^\gamma) \longrightarrow \frac{C_{2,\gamma}}{C_\gamma} - \left(\frac{C_{1,\gamma}}{C_\gamma}\right)^2. \tag{C4}$$

All reference formulas replace C_γ by $C_{\mathrm{col},\gamma}$. No uniformity in entropy order approaching zero or one, or in an unbounded family of moments, is asserted.

To take either endpoint in τ, set $\xi = w(s)u/\tau^{1/\gamma}$. The heat matrix becomes

$$f_{(2|u|)^\gamma,\gamma}\big(T(w(s)|u|/\tau^{1/\gamma})\big).$$

Its eigenvalues are bounded by $\mathrm{e}^{-(2|u|)^\gamma}$ since $\|T(q)\| \leq 1$. This is an integrable common majorant independent of s and τ; weighted energy tests have the analogous majorant $C_p \mathrm{e}^{-(2|u|)^\gamma/2}$. For $u \neq 0$, distinct heights give $T(w|u|/\tau^{1/\gamma}) \to \mathrm{diag}\,\alpha_i$ as $\tau \downarrow 0$. As $\tau \uparrow \infty$ the same matrix tends to P, and continuity of the spectral function at zero includes its kernel. Dominated convergence applies to traces and matrix trace norms, with the extra spatial weight $w(s)$ integrating to one. Writing $A_{\alpha,\gamma}(u) = \mathrm{diag}(\mathrm{e}^{-(2|u|/\alpha_i)^\gamma})$ and $b_\gamma(u) = \mathrm{e}^{-(2|u|)^\gamma}$, this proves

$$\frac{C_\gamma(\tau)}{C_{\mathrm{col},\gamma}(\tau)} \longrightarrow 1, \tag{C5}$$

$$p_{Q,\gamma}(\tau) \longrightarrow 1 - \sum_i \alpha_i^2,$$

$$D_\gamma(\tau) \longrightarrow D^*_{\alpha,\gamma} := \frac{1}{G_\gamma}\int_0^\infty \mathrm{tr}\,|A_{\alpha,\gamma}(u) - b_\gamma(u)P|\,\mathrm{d}u \tag{C6}$$

$$(\tau \downarrow 0), \tag{C7}$$

$$\frac{C_\gamma(\tau)}{C_{\mathrm{col},\gamma}(\tau)} \longrightarrow 1, \tag{C8}$$

$$p_{Q,\gamma}(\tau) \longrightarrow 0, \tag{C9}$$

$$D_\gamma(\tau) \longrightarrow 0 \tag{C10}$$

$$(\tau \uparrow \infty). \tag{C11}$$

For normalization, the integral over $\mathbb{R}$ of the ith diagonal high-resolution heat is $\alpha_i G_\gamma$, while the reference integral is G_γ. Thus the normalized trace-distance prefactor is $1/(2G_\gamma)$ on the full line, or $1/G_\gamma$ on the positive half-line, as written. Positive limiting denominators and the common majorant justify their replacement inside the integral.

The same substitution with $x = (2u/\alpha_i)^\gamma$ gives

$$\frac{C_{p,\gamma}(\tau)}{C_\gamma(\tau)} \longrightarrow \frac{\Gamma(p + 1/\gamma)}{\Gamma(1/\gamma)} \quad (\tau \downarrow 0). \tag{C12}$$

Indeed the positive-half-line integral of $x^p \mathrm{e}^{-x}$ is $\alpha_i \Gamma(p + 1/\gamma)/(2\gamma)$, and the weights sum to one. The dimensionless mean and variance in (C4) both tend to $1/\gamma$. From (C2), the partition ratio at τ and at fixed $r\tau$, and (C3), one obtains

$$\begin{aligned} \lim_{\tau\downarrow 0}\lim_{h\downarrow 0}[S_r(\rho_h) - S_r(\rho_{0,h})] = 0, \\ \lim_{\tau\downarrow 0}\lim_{h\downarrow 0}[S(\rho_h) - S(\rho_{0,h})] = 0. \end{aligned} \tag{C13}$$

These are additive entropy differences. The limits do not merely compare two divergent entropies by their ratio.

For equal weights $\alpha_i = 1/M$, put $a(u) = \mathrm{e}^{-(2Mu)^\gamma}$ and $b(u) = \mathrm{e}^{-(2u)^\gamma}$ on $u > 0$. Since $a \leq b$, the eigenvalues

of $aI_M - bP$ give $\operatorname{tr}|aI_M - bP| = b + (M-2)a$. The integrals of b and a are respectively $G_\gamma/2$ and $G_\gamma/(2M)$. Hence

$$\lim_{\tau\downarrow 0}\lim_{h\downarrow 0}\frac{1}{2}\left\|\rho_h - \rho_{0,h}\right\|_1 = \lim_{\tau\downarrow 0}\lim_{h\downarrow 0}\operatorname{Tr}(Q\rho_h) = 1 - \frac{1}{M}. \tag{C14}$$

The binary decision "multilayer on a Q click, collapsed otherwise" has equal-prior error $(1 - \operatorname{Tr}(Q\rho_h))/2$, since the reference never clicks. Its iterated limit is $1/(2M)$, equal to the single-copy Helstrom optimum from (C14). For any fixed finite number N of independent copies its any-click error tends to $1/(2M^N)$; collective optimality is not needed for this last statement. For unequal weights the Q limit $1 - \sum_i \alpha_i^2$ remains a valid lower bound on the distance, but need not equal the full matrix absolute-value integral.

For completeness that unequal-weight integral has a scalar characterization. The rank-one subtraction $A_{\alpha,\gamma}(u) - b_\gamma(u)P$ has exactly one negative eigenvalue $-r(u)$ for $u > 0$, where $r(u) > 0$ is the unique solution of

$$b_\gamma(u)\sum_i \frac{\alpha_i}{\mathrm{e}^{-(2u/\alpha_i)^\gamma} + r(u)} = 1.$$

The left side at $r = 0$ is strictly larger than one for $M \geq 2$; it decreases continuously to zero. The rank-one eigenvector equation gives the displayed condition and shows that there are no other negative eigenvalues. The signed trace integrates to zero, so $D^*_{\alpha,\gamma} = (2/G_\gamma)\int_0^\infty r(u)\mathrm{d}u$. In particular $1 - \sum_i \alpha_i^2 \leq D^*_{\alpha,\gamma} < 1$, the strict upper bound following from positive definiteness and the strict trace-norm inequality used above.

Every finite-accuracy consequence has the same quantifiers. First choose a finite profile, an accuracy, and a finite list of entropy orders, energy moments, and effects. Choose a sufficiently small τ for their limiting functions, and then a sufficiently small h at that fixed τ. The displayed agreements and distinctions then hold simultaneously to that accuracy. A large but fixed M can be selected before these limits. This gives no numerical onset or uniform prescription for $M(h)$ or $\tau(h)$.

Finally, a physical bounded Lipschitz multiplication detector q on the tube pulls back to $\operatorname{diag}(q(T_{i,h}(s)))$. It differs in norm from $q(\mathbf{r}(s))I_M$ by at most $h\max_i|a_i|\operatorname{Lip}(q)$. Smooth spatial insertions in (A29), followed by the same endpoint substitution, give the common limiting tangential density $w(s)\mathrm{d}s$ at both endpoints; uniform approximation extends this to continuous bounded spatial tests on the curve. The large Q distinction requires coherent comparison of sheet amplitudes in the declared alignment. Neither that unitary identification nor the one-excitation entropy calculation supplies a common laboratory vacuum, an equilibration mechanism, or a many-oscillator thermodynamic assertion.

Appendix D: Controlled preparation of the actual one-excitation states

Throughout this section the curve, its interior tube, the positive smooth density w, the distinct layer heights a_i, their positive weights α_i, and the finite layer number M are fixed. We normalize $\int_\Gamma w\,\mathrm{d}s = \sum_i \alpha_i = 1$ and use

$$\mathcal{H} = L^2(\Gamma,\mathrm{d}s)\otimes\mathbb{C}^M,$$
$$v = (\sqrt{\alpha_1},\dots,\sqrt{\alpha_M})^T,$$
$$P = vv^*,$$
$$Q = I - P.$$

Here and below P, Q also denote their tensor extensions to $\mathcal{H}$. Let R_ε be the actual positive Green inverse after conjugation from the pushforward masses $\alpha_i w(s)\mathrm{d}s$ on $T_{i,\varepsilon}(s) = \gamma(s) + \varepsilon a_i n(s)$. The reference is $R_0 = R_{\mathrm{base}}\otimes P$. The physical eigenvalues of $L_h = R_h^{-1}$, $h \in \{0,\varepsilon\}$, are $\lambda_j(L_h) = r_j(R_h)^{-1}$, where r_j lists the positive eigenvalues in decreasing order. The reference alignment complement is assigned heat zero; it is never inserted as a set of zero-energy modes. Here ε is the geometric separation denoted by h in the main text; the local index $h \in \{0,\varepsilon\}$ only selects the reference or split candidate in this preparation section. All estimates below are uniform only for sufficiently small $\varepsilon > 0$ with these fixed data.

1. An actual-operator weak trace-ideal estimate

We prove, including the cross-layer coupling, that

$$r_j(R_h) \leq \frac{K}{j}, \qquad \lambda_j(L_h) \geq cj, \qquad c = K^{-1} > 0. \tag{D1}$$

Thus R_h is uniformly in the weak trace ideal $\mathfrak{S}_{1,\infty}$. The assertion for $h = 0$ concerns only its nonzero physical spectrum.

a. Uniform diagonal regularity. Let ℓ be the length of Γ, parametrized by arclength. On a fixed closed interval $0 \leq \varepsilon \leq \varepsilon_*$, every individual offset remains embedded, has speed bounded away from zero, and has uniformly bounded derivatives. Its diagonal layer kernel is

$$R_{ii,\varepsilon}(s,t) = \alpha_i\sqrt{w(s)w(t)}G(T_{i,\varepsilon}(s),T_{i,\varepsilon}(t)).$$

The regular part of the Dirichlet Green function is smooth on the tube product, also with any fixed distant Dirichlet masks. The quotient

$$\frac{|T_{i,\varepsilon}(s) - T_{i,\varepsilon}(t)|}{(\ell/\pi)|\sin(\pi(s-t)/\ell)|}$$

extends positively and smoothly across $s = t$ as a periodic function. Indeed, near the diagonal division of the chord by $s - t$ gives the integral of the nonvanishing tangent; division of the squared norms followed by the positive square root proves smoothness. Away from the

diagonal compactness and uniform embeddedness give a positive lower bound. All fixed orders of derivatives are uniformly bounded. Subtracting $-(2\pi)^{-1}\log|2\sin(\pi(s-t)/\ell)|$ therefore leaves a uniformly smooth family. For the nonzero Fourier frequencies $k_n = 2\pi n/\ell$, this logarithmic convolution has multiplier $1/(2|k_n|)$. Its zero multiplier is bounded and immaterial to the estimate. It maps L^2 to H^1 boundedly, as do the smooth remainders and multiplication by $\sqrt{w}$. Consequently

$$\sup_{i,\varepsilon} \|R_{ii,\varepsilon} : L^2 \longrightarrow H^1\| < \infty.$$

The embedding $H^1(\Gamma) \hookrightarrow L^2(\Gamma)$ has singular values bounded by C/j, by the same Fourier basis. Factoring each diagonal block through this embedding yields $s_j(R_{ii,\varepsilon}) \le C_d/j$ with one common constant. The base curve obeys the same estimate.

b. The cross blocks are controlled by positivity. For every positive operator R on a sum of M Hilbert spaces, let E_i inject the ith summand. Then

$$\begin{aligned}\langle f, Rf\rangle &= \left\|\sum_i R^{1/2}E_i f_i\right\|^2 \\ &\le M\sum_i \|R^{1/2}E_i f_i\|^2 = M\sum_i \langle f_i, R_{ii} f_i\rangle.\end{aligned}$$

Hence, as a form inequality of bounded positive operators,

$$0 \le R_\varepsilon \le M\ \mathrm{diag}(R_{11,\varepsilon},\dots,R_{MM,\varepsilon}). \tag{D2}$$

Each diagonal block is positive because it is a compression of R_ε. The number of its eigenvalues exceeding $t>0$ is at most C_d/t. The direct sum consequently has at most MC_d/t such eigenvalues. Min–max applied to (D2) proves (D1), after enlarging a fixed constant to include the reference. No uniform smoothness of a cross-layer kernel was assumed. Such a smoothness assertion would fail as the layers collide. No constant uniform in growing M or vanishing weights is claimed.

2. Heat tails and a common tensor encoding

Fix $\tau,\gamma>0$, set $\beta=\tau\varepsilon^\gamma$, and write $A_h=\exp(-\beta L_h^\gamma)$ with the reference convention above. Retain its first J physical spectral terms in $A_{h,J}$, and put $Z_h=\operatorname{Tr}A_h$, $Z_{h,J}=\operatorname{Tr}A_{h,J}$ and $T_{h,J}=Z_h-Z_{h,J}$. For $a=\tau c^\gamma$, monotonicity and an integral comparison give the explicit all-mode estimate

$$\begin{aligned}T_{h,J} &\le \sum_{j>J}\exp[-a(\varepsilon j)^\gamma] \le \frac{1}{\varepsilon}\int_{\varepsilon J}^\infty \exp(-ax^\gamma)\,\mathrm{d}x \\ &= \frac{\Gamma(1/\gamma, a(\varepsilon J)^\gamma)}{\varepsilon\gamma a^{1/\gamma}}.\end{aligned} \tag{D3}$$

Here $\Gamma(s,y)=\int_y^\infty t^{s-1}e^{-t}\mathrm{d}t$. For fixed $s>0$ and $y\ge 1$ it is at most $B_s e^{-y/2}$: bound $t^{s-1}e^{-t/2}$ on $[1,\infty)$ and integrate the remaining exponential. The actual joint heat law, Eq. (A8), gives constants $0<z_*\le z^*<\infty$ such that, eventually, under either hypothesis,

$$\frac{z_*}{\varepsilon} \le Z_h \le \frac{z^*}{\varepsilon}. \tag{D4}$$

For any $0<\delta<1$, equations (D3) and (D4) allow

$$J=\left\lceil\frac{\kappa_\delta}{\varepsilon}\right\rceil, \qquad \kappa_\delta = O\big([1+\log(1/\delta)]^{1/\gamma}\big), \qquad \frac{T_{h,J}}{Z_h}\le\delta. \tag{D5}$$

The retained and discarded spectral supports are orthogonal, so

$$\frac{1}{2}\left\|\frac{A_h}{Z_h}-\frac{A_{h,J}}{Z_{h,J}}\right\|_1 = \frac{T_{h,J}}{Z_h}\le\delta. \tag{D6}$$

This is an asymptotic existence estimate with no certified finite onset.

Let $W\subset L^2(\Gamma,\mathrm{d}s)$ be the span of all M scalar components of the J retained target eigenvectors and the J retained base-curve eigenfunctions. Then $n=\dim W\le (M+1)J$ and

$$\mathcal{S}=W\otimes\mathbb{C}^M, \qquad D=Mn\le M(M+1)J. \tag{D7}$$

Both retained state ranges lie in $\mathcal{S}$, which reduces every fixed transverse matrix. For an orthonormal-basis isometry $V_W:\mathbb{C}^n\to W$, set $V=V_W\otimes I_M$. Thus

$$\begin{aligned}\rho_{D,h} &= \frac{V^*A_{h,J}V}{Z_{h,J}}, \\ Q_D &= V^*QV = I_n\otimes Q, \\ Q_D\rho_{D,0} &= 0.\end{aligned} \tag{D8}$$

The eigenbases of the two actual candidates need not agree. For an original physical spatial multiplication effect M_f, its encoded effect is precisely V^*M_fV, generally a dense matrix. The physical spatial measurement class remains this compressed family; arbitrary output-port effects do not acquire spatial locality by being assigned a coordinate label. A fixed layer unitary sending v to one output implements Q_D in each of the n longitudinal modes. Combining one known M-component vector uses $M-1$ two-mode rotations, hence $O(D)$ elements overall; the physical conversion into the modes W, routing, and transmission have separate costs.

a. Weighted tails, rather than an entropy continuity shortcut. Let $y_j=\beta\lambda_j^\gamma$, $q=T_J/Z$, and $m=Z^{-1}\sum_j y_j e^{-y_j}$. With m_J its retained normalized version and $m_{\mathrm{tail}}=Z^{-1}\sum_{j>J}y_je^{-y_j}$,

$$\begin{aligned}S(\rho)-S(\rho_J) &= -\log(1-q)+m-m_J, \\ |m-m_J| &\le \frac{m_{\mathrm{tail}}+qm}{1-q}.\end{aligned}$$

For each fixed $p\ge 0$, $y^pe^{-y}\le C_pe^{-y/2}$, so (D3) at $\tau/2$ controls each chosen moment tail. The weighted joint

law bounds the full moments. For any selected positive Rényi order $r \neq 1$, use also the cutoff estimate at $r\tau$ and the identity $S_r = (\log Z(r\beta) - r \log Z(\beta))/(1-r)$. Thus a fixed finite list of thermal scalars has controlled truncation error with the same dimension order, without an extra $\log(1/\varepsilon)$ factor. Arbitrary preparation errors small only in trace distance do not imply this stronger assertion at growing dimension or for unbounded energy observables.

3. Deliberate single-excitation preparation and its costs

On the common encoded space let $A_{D,h} = V^* A_{h,J} V$ and $F_h = A_{D,h}^{1/2}$. Inject a heralded excitation into a uniformly random input port, giving I_D/D, and apply the passive contraction F_h. The surviving block, success probability, and conditional state are exactly

$$\sigma_h = \frac{A_{D,h}}{D}, \qquad s_h = \frac{Z_{h,J}}{D}, \qquad \frac{\sigma_h}{s_h} = \rho_{D,h}. \tag{D9}$$

In particular,

$$s_h \geq \frac{(1-\delta) z_*}{M(M+1)(\kappa_\delta + \varepsilon)}. \tag{D10}$$

At each fixed accuracy this is a positive lower bound independent of small ε. It is preparation of an injected excitation, not selection of one quantum from an equilibrium Bose sample. In that different protocol the probability is Z_1/Z_{B} and can be exponentially small on the joint scale.

The exact dilation

$$U_F = \begin{pmatrix} F & (I-F^2)^{1/2} \\ (I-F^2)^{1/2} & -F \end{pmatrix}$$

is unitary because its blocks commute. Vacuum auxiliary ports realize F on the surviving one-excitation sector. Generic universal compilation takes $O(D^2)$ two-mode elements and $O(D)$ depth [14]. Those counts do not establish constant transmission: fixed loss per traversal at depth proportional to D may make the collection probability exponentially small.

An alternative is to draw a retained eigenlabel with its normalized Gibbs probability and prepare that known encoded vector. For $c \in \mathbb{C}^D$, pad to $D' = 2^d$, $d = \lceil \log_2 D \rceil$. At each node of a binary splitter tree, split power according to the two descendant sums of $|c_i|^2$; at the leaves impose phases $\arg c_i$. Induction gives the desired vector with $D' - 1 = O(D)$ splitters and depth d. Under the explicit assumption of a common intensity transmission $\vartheta \in (0,1]$ per level, balanced paths and calibrated phases, conditioning on survival preserves the vector and

$$q_{\mathrm{tree}} \geq \vartheta^{\lceil \log_2 D \rceil} \geq \vartheta D^{-\kappa}, \qquad \kappa = -\frac{\log \vartheta}{\log 2}.$$

A mere lower bound on unequal path transmissions would not preserve the vector. If known label-dependent transmissions q_j preserve each individual vector, sample label j with probabilities proportional to p_j/q_j. Its accepted mixture then has weights p_j and acceptance $(\sum_j p_j/q_j)^{-1} \geq \min_j q_j$. The eigenvectors, their certified approximation, $O(DJ)$ dense coefficient storage, $O(D)$ settings per reconfiguration, settling and reset are paid resources. This proves a polynomial transmission budget only under the stated balanced-loss and polynomial-control assumptions. It is neither an efficient continuum eigensolver nor an unknown-specimen oracle: a programmed matrix is already known, whereas a specimen experiment requires physical access to the unknown generator.

For N accepted copies, source delivery and state-independent collection factors $q_{\mathrm{src}}, q_{\mathrm{col}}$ yield mean raw attempt count $N/(s_h q_{\mathrm{src}} q_{\mathrm{col}})$. Failures, source reset and filtering time remain charged. A final destructive click can certify a completed measured trial; availability of a stored excitation requires an actual nondestructive herald. A failed arrival is not a measurement of $I - Q$. Acceptance rates, input labels, waiting times, or resolved failure channels may themselves reveal the hypothesis; claims about conditional normalized copies are not bounds on an enlarged experiment retaining those records.

4. Square-root Gibbs weights from ordinary loss

For $\gamma = 1/2$ the classical subordination identity [22] is

$$\begin{aligned} e^{-\beta\sqrt{L}} &= \int_0^\infty \ell_\beta(u) e^{-uL}\, \mathrm{d}u, \\ \ell_\beta(u) &= \frac{\beta}{2\sqrt{\pi}\, u^{3/2}} e^{-\beta^2/(4u)}. \end{aligned} \tag{D11}$$

For completeness, set $I(b) = \int_0^\infty e^{-x^2 - b^2/x^2} \mathrm{d}x$ for $b > 0$. Differentiation under the integral and the substitution $x = b/y$ give $I'(b) = -2I(b)$. Since $I(0) = \sqrt{\pi}/2$ by dominated convergence, $I(b) = (\sqrt{\pi}/2) e^{-2b}$. The substitution $x = \beta/(2\sqrt{u})$ proves the scalar Laplace identity in (D11); the spectral theorem proves its operator form. Taking the scalar spectral value zero proves that ℓ_β integrates to one.

Suppose a finite positive supported matrix L has accessible linear loss jumps $J_\mu = \sum_i b_{\mu i} a_i$ with $B^*B = L$. Under the Lindblad convention

$$\dot\rho = \sum_\mu \left(J_\mu \rho J_\mu^* - \tfrac{1}{2}\{J_\mu^* J_\mu, \rho\} \right),$$

every jump from the one-excitation sector enters vacuum. Its surviving block therefore evolves as $\rho_{\mathrm{surv}}(u) = e^{-uL/2} \rho_{\mathrm{surv}}(0) e^{-uL/2}$, the standard conditional-loss evolution [23]. Apply this with L the restriction of the actual L_h to its J retained physical eigenvectors. Project the input I_D/D onto that support, counting its unused

coordinates as initial failures. For sampled duration u the unnormalized survivor is e^{-uL}/D. Pooling these unnormalized survivors with density ℓ_β gives $e^{-\beta\sqrt{L}}/D$ and hence exactly the state and success in (D9). Normalizing at each duration before averaging would change the preparation. A commuting coherent drift leaves the diagonal surviving block unchanged; a general noncommuting drift needs its own control estimate.

a. A finite maximum duration. The sampled duration has infinite mean. Abort proposed draws $u > U$ before injection, or resample them classically. Their probability is

$$q_U = \operatorname{erf}\left(\frac{\beta}{2\sqrt{U}}\right) \le \frac{\beta}{\sqrt{\pi U}}. \tag{D12}$$

Put $A = e^{-\beta\sqrt{L}}$, $A_U = \int_0^U \ell_\beta(u)e^{-uL}\mathrm{d}u$ and $b = \operatorname{Tr}(A - A_U)$. Then $0 \le A - A_U \le q_U I_J$ and $b \le Jq_U \le Dq_U$. Adding and subtracting $A_U/\operatorname{Tr} A$ gives

$$\frac{1}{2}\left\|\frac{A}{\operatorname{Tr} A} - \frac{A_U}{\operatorname{Tr} A_U}\right\|_1 \le \frac{b}{\operatorname{Tr} A} \le \frac{q_U}{s}, \qquad s = \frac{\operatorname{Tr} A}{D}. \tag{D13}$$

If $s_* \le s$ and additional state distance $\delta_p \in (0,1)$ is allowed, choose

$$U \ge \frac{\beta^2}{\pi\delta_p^2 s_*^2}. \tag{D14}$$

Counting all proposed draws gives success at least $(1-\delta_p)s$. Resampling before injection gives per-injection success $\operatorname{Tr} A_U/[D(1-q_U)]$ and mean classical draw count $1/(1-q_U)$. Every injected filtering run lasts at most U. At $\beta = \tau\sqrt{\varepsilon}$ and fixed accuracies the chosen U is $O(\varepsilon)$ in the mathematical clock of L.

b. Exact finite mean with first-loss stopping. Alternatively, assume all losses are promptly and reliably monitored. Sample the full duration distribution but stop at the first loss or the sampled endpoint. In input eigenmode j, the loss time is exponential with rate λ_j, independently of the sampled duration. Tonelli's theorem and its Laplace transform give

$$\mathbb{E}\min(u, T_j) = \int_0^\infty \Pr(u > t)e^{-\lambda_j t}\mathrm{d}t = \frac{1 - e^{-\beta\sqrt{\lambda_j}}}{\lambda_j}.$$

Including initial support failures with zero filtering time,

$$\begin{aligned}\mathbb{E}T_{\text{attempt}} &= \frac{1}{D}\sum_{j\le J}\frac{1 - e^{-\beta\sqrt{\lambda_j}}}{\lambda_j} \le \frac{\beta}{D}\sum_{j\le J}\lambda_j^{-1/2} \\ &\le \frac{2\beta\sqrt{J}}{D\sqrt{c}} = O(\varepsilon),\end{aligned} \tag{D15}$$

because $D \ge J$ and (D5) holds. The renewal equation gives mean total filter time until acceptance $\mathbb{E}T_{\text{attempt}}/s$; time and success need not be independent within a trial. The accepted state remains exact on the supported spectrum. This supplies a mean, not a maximum, and assumes prompt jump registration, controlled false aborts and finite resets. Input support sorting and reset times are additional costs.

5. The physical clock, locality, and finite reference support

Let $\Lambda = \max_{j\le J}\lambda_j$. With maximal implementable loss rate Γ_{hw} use $L_{\text{ph}} = (\Gamma_{\text{hw}}/\Lambda)L$, so that

$$t_{\text{ph}} = \frac{\Lambda}{\Gamma_{\text{hw}}}u. \tag{D16}$$

The needed upper bound $\Lambda = O_\delta(\varepsilon^{-1})$ does not follow from the lower spectral bound (D1). It follows instead from the actual counting law, Eq. (B5): its coefficient satisfies $\mathcal{N}_\alpha(E)/E \to 1/(2\pi)$ as $E \to \infty$. Indeed the local matrix $S(q)$ tends to $\operatorname{diag}(\alpha_i)$, its inverse-energy branches obey $e_i(s,q) \sim 2q/[w(s)\alpha_i]$, and their radial crossings satisfy $q_i(s,E)/E \to w(s)\alpha_i/2$, uniformly in s. Integrate and use the two angular signs and total mass one. Choose a fixed E with $\mathcal{N}_\alpha(E) > \kappa_\delta$, and similarly for the base reference. The strict margin in the actual limit $\varepsilon\#\{\lambda_j \le E/\varepsilon\} \to \mathcal{N}_\alpha(E)$ pays the ceiling in J. Thus (D14) gives a bounded maximum physical filter time $O_{\delta,\delta_p,\tau}(\Gamma_{\text{hw}}^{-1})$, and (D15) gives a bounded mean with monitoring. This does not provide a numerical onset, a uniform simultaneous $\delta(\varepsilon) \to 0$ estimate, or a size-independent realizable Γ_{hw}.

If a mass-normalized graph stiffness actually has

$$\begin{gathered}K = \sum_{ij} g_{ij}(e_i - e_j)(e_i - e_j)^* + \sum_i g_{i0}e_i e_i^*, \\ L = M_b^{-1/2}KM_b^{-1/2}, \\ M_b = \operatorname{diag}(m_i) > 0,\end{gathered}$$

the jumps $\sqrt{g_{ij}}(a_i/\sqrt{m_i} - a_j/\sqrt{m_j})$ and $\sqrt{g_{i0}/m_i}\,a_i$ implement that L. Their neighboring modes require a common loss reservoir. For the actual curve operator, eliminating the carrier gives a Dirichlet-to-Neumann map, or on a mesh the Schur complement

$$\begin{aligned}K_{\text{eff}} &= K_{BB} - K_{BI}K_{II}^{-1}K_{IB}, \\ L_{\text{eff}} &= M_B^{-1/2}K_{\text{eff}}M_B^{-1/2}.\end{aligned}$$

This is generally dense even if the uneliminated carrier is local [24]. Static elimination does not establish that ordinary local bulk damping produces $e^{-uL_{\text{eff}}}$ on the boundary. Positive bulk inertia adds dynamical modes and a frequency-dependent elimination. A specimen construction must prove its common mode map, generator approximation, heat and weighted-tail errors, and resources across the growing band.

When λ_j denotes squared mechanical frequency, a retained single excitation has maximum frequency $O_\delta(\varepsilon^{-1/2})$ and its energy, in fixed physical units, must be supplied and reset. An optical path encoding at a fixed photon carrier reproduces the encoded state; that carrier

energy is not the mechanical excitation energy. Native mechanical oscillators with different frequency matrices generally have different annihilation operators and vacua, related by a Bogoliubov transformation. Second quantization of the mathematical L^2 alignment specifies a common number-conserving Fock model; it does not prove a passive, vacuum-preserving laboratory identification.

A valid continuum soft reference retaining longitudinal coercivity is

$$H_{\mathrm{soft}} = L^{\gamma}_{\mathrm{base}} \otimes I_M + \Delta I \otimes Q.$$

Its one-excitation Gibbs weight on Q and its trace distance from the hard reference are both

$$p_{Q,\mathrm{soft}} = \frac{(M-1)e^{-\beta\Delta}}{1+(M-1)e^{-\beta\Delta}}. \tag{D17}$$

Hence error at most δ_{ref} follows from $\beta\Delta \geq \log[(M-1)(1-\delta_{\mathrm{ref}})/\delta_{\mathrm{ref}}]$ when this is positive. A constant penalty on an otherwise dispersionless infinite Q space would have infinite thermal degeneracy. Also a spring changes squared frequencies; it does not automatically realize the displayed additive frequency-Hamiltonian penalty. Full Bose vacuum-reference accuracy has a separate population budget, as discussed with the Bose model.

a. Error and resource bookkeeping. If the continuum truncation distance is δ, the additional preparation distance is δ_p, and the effect error is at most δ_Q in operator norm, the measured Q probability differs from its continuum value by at most $\delta + \delta_p + \delta_Q$. For two contractions $F, \widetilde{F}$ with $\|\widetilde{F} - F\| \leq d_F$,

$$\left\| \frac{\widetilde{F}\widetilde{F}^* - FF^*}{D} \right\|_1 \leq 2d_F, \qquad D_{\mathrm{tr}}(\widetilde{\rho}, \rho) \leq \frac{2d_F}{s},$$

provided the perturbed successful output is nonzero. The first bound follows by expanding the difference into two products and using $\|X\|_1 \leq D\|X\|$; the second follows by comparing normalizations. This is a whole-device norm tolerance, not a constant allowance on each of $O(D^2)$ optical elements. Mode-dependent damping changes the state. Even an extra scalar damping rate reweights the random durations in the pooled subordination experiment and cannot generally be removed as one overall transmission number. A sufficient small-loss condition over a physical shot of duration T is $\max_j \kappa_j T \leq \delta_{\mathrm{loss}}$; for $\kappa_j = \omega_j/Q_{\mathrm{mech},j}$ this requires $Q_{\mathrm{mech},j} \geq \omega_j T/\delta_{\mathrm{loss}}$. Preparation and measured-copy cost therefore includes geometry and coefficient computation, compilation and calibration, followed by

$$\frac{N}{s q_{\mathrm{src}} q_{\mathrm{col}}} \big(\text{injection} + \text{filter time} + \text{reset} + \text{readout}\big),$$

with a renewal expectation for variable-duration trials and actual conditional transmission where it is state dependent. Storage, number resolution, dark counts, additional mode conversion and control updates are separate resources. Only the ideal acceptance floor and the displayed conditional filter-time estimates are proved here.

Appendix E: Full Bose ensembles: counting, thermodynamics and receiver costs

The normalized one-excitation heat state and the full bosonic equilibrium state are different statistical experiments. This appendix constructs the second experiment in a specified common number-conserving representation, proves its counting and thermodynamic laws, and identifies the resources behind its discrimination bounds. Its physical premise is a common annihilation-mode encoding. The geometric L^2 identification alone does not establish that premise.

1. The ensemble and the spectral estimates used below

Fix the curve, carrier and permitted distant masks, the smooth positive density w with $\int_\Gamma w\,\mathrm{d}s = 1$, and a finite profile with $M \geq 2$ distinct heights a_j and weights $\alpha_j > 0$, $\sum_j \alpha_j = 1$. All these data, the exponent $\gamma > 0$ and $\tau > 0$ are fixed before $h \downarrow 0$. Set

$$\begin{aligned} \mathfrak{h} &= L^2(\Gamma, \mathrm{d}s) \otimes \mathbb{C}^M, \\ v &= (\sqrt{\alpha_j})_{j=1}^M, \\ P &= vv^*, \\ Q &= I - P, \\ \beta &= \tau h^{\gamma}. \end{aligned} \tag{E1}$$

We also use P, Q for their tensor extensions to $\mathfrak{h}$. Write $R_{1,h} = R_h$ for the actual positive Green operator and $L_{1,h} = R_h^{-1}$ for its positive mass eigenvalue operator. The collapsed physical Green operator is $R_{0,h} = \widetilde{R}$ on $L^2(\Gamma)$, independent of h, and $L_{0,h} = L_0 = \widetilde{R}^{-1}$. The map $f \mapsto f \otimes v$ identifies this scalar physical space with $P\mathfrak{h}$. An inverse power of the aligned collapsed operator is used only through its heat, extended by zero on $Q\mathfrak{h}$. In particular define

$$\begin{aligned} \Theta_{1,h} &= A_h = \mathrm{e}^{-\beta L_{1,h}^{\gamma}}, \\ \Theta_{0,h} &= A_{0,h} = \mathrm{e}^{-\beta L_0^{\gamma}} \otimes P, \\ N_{i,h} &= \Theta_{i,h}(I - \Theta_{i,h})^{-1}. \end{aligned} \tag{E2}$$

Thus $\Theta_{1,h}, \Theta_{0,h}$ are local aliases for the heat operators $A_h, A_{0,h}$, whereas the scaled Green operator in the geometric appendix is $X_h = R_h/h$. The index $i = 0$ always refers to the physical collapsed spectrum with a vacuum extension on Q.

The uniform spectral bound proved in (D1) gives constants $c, C > 0$, independent of sufficiently small h, such that the nonzero physical eigenvalues satisfy

$$\begin{aligned} r_j(R_{i,h}) &\leq Cj^{-1}, \\ \lambda_j(L_{i,h}) &\geq cj, \\ j &\geq 1, \\ i &= 0, 1. \end{aligned} \tag{E3}$$

There are no additional zero-energy eigenvalues on the reference complement. Recall why the estimate is uniform: each diagonal self-curve Green block is a weighted periodic logarithmic operator plus a uniformly smooth remainder and maps L^2 to H^1 uniformly. Its singular values are $O(j^{-1})$. Positivity of the full block operator gives $0 \le R_h \le M\,\mathrm{diag}(R_{11,h},\dots,R_{MM,h})$ by applying $\|\sum_{j=1}^M y_j\|^2 \le M\sum_j \|y_j\|^2$ to $y_j = R_h^{1/2} I_j x_j$. Min–max and the fixed finite direct sum give (E3). Thus this argument never requires off-diagonal collision kernels to remain uniformly smooth.

For any $t>0$, decreasing-integral comparison yields

$$\mathrm{Tr}\,\mathrm{e}^{-tL_{i,h}^{\gamma}} \le \sum_{j\ge 1} \mathrm{e}^{-t(cj)^{\gamma}} \le \int_0^{\infty} \mathrm{e}^{-t(cx)^{\gamma}}\,\mathrm{d}x \le K_\gamma t^{-1/\gamma}, \tag{E4}$$

$$h\,\mathrm{Tr}\Big[(k\beta L_{i,h}^{\gamma})^p \mathrm{e}^{-k\beta L_{i,h}^{\gamma}}\Big] \le K_{p,\gamma}(k\tau)^{-1/\gamma}, \tag{E5}$$

$$k \ge 1, \tag{E6}$$

$$p = 0,1,2. \tag{E7}$$

The second estimate follows for $p>0$ from $x^p\mathrm{e}^{-x} \le c_p \mathrm{e}^{-x/2}$; all reference traces are on its physical spectrum. For each fixed $h>0$, $\Theta_{i,h}$ is positive trace class with norm less than one. Therefore $N_{i,h}$ is positive trace class. The same lower spectral bound implies

$$\|N_{i,h}\| \le Ch^{-\gamma}, \qquad \nu_j(N_{i,h}) \le Ch^{-\gamma}j^{-\gamma}, \tag{E8}$$

where ν_j are the decreasing nonzero eigenvalues. These estimates, not the bounded heat-test theorem alone, control the Bose singularity at small dimensionless energy.

For a contraction T on $\mathfrak{h}$, let $\Gamma_{\mathrm{s}}(T) = \bigoplus_{n\ge 0} T^{\otimes_{\mathrm{s}} n}$, with value one on the vacuum sector. On $\mathcal{F}_{\mathrm{s}}(\mathfrak{h})$, define

$$\mathcal{Z}_{i,h} = \det(I-\Theta_{i,h})^{-1}, \qquad \varrho_{i,h} = \det(I-\Theta_{i,h})\Gamma_{\mathrm{s}}(\Theta_{i,h}). \tag{E9}$$

Diagonalizing Θ with eigenvalues $0 \le \theta_j < 1$ gives $\mathrm{Tr}_{\mathcal{F}}\Gamma_{\mathrm{s}}(\Theta) = \prod_j (1-\theta_j)^{-1}$. The product is finite because $\sum_j -\log(1-\theta_j) \le \mathrm{Tr}\,\Theta/(1-\|\Theta\|)$. This proves existence and normalization, including the infinite number of physical modes. The signal has excitation Hamiltonian $\mathcal{H}_{1,h} = \mathrm{d}\Gamma(L_{1,h}^{\gamma})$. The reference is the Gibbs state of $\mathrm{d}\Gamma(L_0^{\gamma})$ on $\mathcal{F}_{\mathrm{s}}(P\mathfrak{h})$, tensored with exact vacuum on $\mathcal{F}_{\mathrm{s}}(Q\mathfrak{h})$. It can be described as exclusion of that sector at infinite energy; it is not a finite-energy Gibbs Hamiltonian on the whole common space. All energies below are excitation energies relative to these stipulated vacua; no zero-point or Casimir energy comparison is included. The isolated Hamiltonian conserves total number, while preparation at zero chemical potential allows a bath to exchange quanta. A fixed-total-number canonical ensemble is a different model.

2. A noncommuting counting determinant

Let E be any orthogonal projection on $\mathfrak{h}$ and $0 \le \eta \le 1$ the independent detection efficiency. The no-detection effect is $\Gamma_{\mathrm{s}}(I-\eta E)$, since its value on m quanta in $E\mathfrak{h}$ is $(1-\eta)^m$. At $\eta = 1$ this means the vacuum projector on that factor, including value one at $m=0$. For either state in (E9), or for any positive trace-class Θ of norm less than one, one has

$$p_{\mathrm{no}}(\eta,E) = \det_{E\mathfrak{h}}(I_E + \eta ENE)^{-1}, \qquad N = \Theta(I-\Theta)^{-1}. \tag{E10}$$

In particular E need not commute with Θ, N or the energy.

To prove the formula, put $T = I-\eta E$. Multiplicativity of second quantization and a positive cyclic sandwich give

$$\begin{aligned}\mathrm{Tr}_{\mathcal{F}}\big[\Gamma_{\mathrm{s}}(\Theta)\Gamma_{\mathrm{s}}(T)\big] &= \mathrm{Tr}_{\mathcal{F}}\,\Gamma_{\mathrm{s}}(\Theta^{1/2}T\Theta^{1/2}) \\ &= \det(I-\Theta+\eta\Theta^{1/2}E\Theta^{1/2})^{-1}.\end{aligned}$$

The cyclic step is legitimate: $\Gamma_{\mathrm{s}}(\Theta^{1/2})$ is Hilbert–Schmidt because its squared Hilbert–Schmidt norm equals $\mathrm{Tr}_{\mathcal{F}}\Gamma_{\mathrm{s}}(\Theta) < \infty$, and $\Gamma_{\mathrm{s}}(T)$ is bounded. The sandwich is positive, trace class and bounded above by Θ, so the partition product applies. Factor the one-particle determinant by $I-\Theta$ to obtain $\det(I+\eta N^{1/2}EN^{1/2})^{-1}$ after normalization. For completeness, the required determinant algebra can first be applied to spectral truncations Θ_m; then $\Theta_m \to \Theta$ and $N_m \to N$ in trace norm, and all displayed positive sandwiches converge in trace norm. Continuity of the Fredholm determinant passes the finite-dimensional identities to the limit. Finally $N^{1/2}E$ and $EN^{1/2}$ are Hilbert–Schmidt and the identity $\det(I+XY) = \det(I+YX)$ moves the determinant to $E\mathfrak{h}$. This proves (E10). The finite-dimensional Fock trace identity is classical [13]; the preceding argument supplies the needed trace-class extension.

The same calculation gives the probability generating function for the detected E-count K:

$$\mathbb{E}z^K = \det_{E\mathfrak{h}}[I_E + \eta(1-z)ENE]^{-1}, \qquad 0 \le z \le 1. \tag{E11}$$

It follows either by conditioning the independent loss on the number in E, or by replacing η in the no-click identity with $\eta(1-z)$.

For $q = |\xi| > 0$, retain the geometric symbol

$$\begin{aligned} B(s,\xi) &= \frac{w(s)}{2q}\big[\sqrt{\alpha_i\alpha_j}\,\mathrm{e}^{-q|a_i-a_j|}\big]_{i,j=1}^{M}, \\ C_{Q,\gamma}(t) &= \frac{1}{2\pi}\int_{\Gamma}\int_{\mathbb{R}} \mathrm{tr}\big(Q\mathrm{e}^{-tB(s,\xi)^{-\gamma}}\big)\,\mathrm{d}\xi\,\mathrm{d}s. \end{aligned} \tag{E12}$$

The heat law (A8), with its fixed matrix insertion, gives $h\,\mathrm{Tr}(Q\Theta_{1,h}) \to C_{Q,\gamma}(\tau) > 0$. Positivity is strict: $B(s,\xi) > 0$ for $\xi \ne 0$, and Q is a nonzero projection. Since $N \ge \Theta$, positive compression, eigenvalue monotonicity and $\log(1+\eta x) \ge x\log(1+\eta)$ for $0 \le x \le 1$ show that

$$\begin{aligned} -\log p_{\mathrm{no}}(\eta,Q) &= \mathrm{Tr}_Q \log(I_Q + \eta QNQ) \\ &\ge \log(1+\eta)\,\mathrm{Tr}(Q\Theta). \end{aligned} \tag{E13}$$

Consequently, for every fixed $0 < \eta \le 1$ and any $0 < c < \log(1+\eta) C_{Q,\gamma}(\tau)$, all sufficiently small h satisfy

$$\begin{aligned} p_{\mathrm{no},1}(\eta, Q) &\le \mathrm{e}^{-c/h}, \\ P_{\mathrm{err}}^{\mathrm{click}} = \tfrac{1}{2} p_{\mathrm{no},1}(\eta, Q) &\le \tfrac{1}{2}\mathrm{e}^{-c/h}, \\ \tfrac{1}{2}\|\varrho_{1,h} - \varrho_{0,h}\|_1 \ge 1 - p_{\mathrm{no},1}(\eta, Q) &\longrightarrow 1. \end{aligned} \tag{E14}$$

The equal-prior rule declares the signal if any transverse quantum is detected. The reference has zero false-positive probability because $QN_{0,h} = 0$. The trace-distance lower bound follows by using this same effect in the variational characterization of trace distance. No claim of finite-h Helstrom optimality is made. For k independent whole preparations and the ideal reference, an any-click-across-gates rule has error $p_{\mathrm{no},1}^k/2$; it consumes k whole thermal samples.

Finite Fourier detection suffices. If $D_s = -\mathrm{i}\partial_s$ on the periodic curve of length ℓ_Γ and $0 < d < R < \infty$ are fixed, let $E_h = \mathbf{1}_{\{d<|hD_s|<R\}} \otimes Q$. Then

$$\operatorname{rank} E_h = \frac{(M-1)\ell_\Gamma(R-d)}{\pi h} + O(1), \tag{E15}$$

$$h \operatorname{Tr}(E_h \Theta_{1,h}) \tag{E16}$$

$$\longrightarrow \frac{1}{2\pi}\int_\Gamma \int_{d<|\xi|<R} \operatorname{tr}(Q\mathrm{e}^{-\tau B^{-\gamma}})\,\mathrm{d}\xi\,\mathrm{d}s > 0. \tag{E17}$$

The second assertion is the sharp Fourier insertion law (A33). Its treatment of the discontinuous band edges uses smooth multiplier insertions, then bounds the two transition strips by their $O(\text{strip width}/h)$ ranks and $0 \le \Theta \le I$. Thus it concerns the actual geometric heat and its complete tails. Applying (E13) to E_h proves an exponential error bound with finitely many, $O(h^{-1})$, measured modes. This rank count does not supply a bandwidth, a mode converter or a simultaneous readout apparatus.

3. All-mode Bose thermodynamics and the price of one sample

For $p = 0, 1, 2$ define the split coefficients

$$\begin{aligned} C_{p,\gamma}(t) &= \frac{1}{2\pi}\int_\Gamma \int_{\mathbb{R}} \operatorname{tr}\big[(tB^{-\gamma})^p \mathrm{e}^{-tB^{-\gamma}}\big]\,\mathrm{d}\xi\,\mathrm{d}s, \\ C_\gamma &= C_{0,\gamma}. \end{aligned} \tag{E18}$$

Reference coefficients will carry a superscript col, to distinguish a collapsed index from the moment order $p = 0$. The actual convergence to these integrals is (C1). Put

$$\begin{aligned} \mathcal{L}_\gamma(\tau) &= \sum_{k\ge1} \frac{C_\gamma(k\tau)}{k}, \\ \mathcal{E}_\gamma(\tau) &= \sum_{k\ge1} \frac{C_{1,\gamma}(k\tau)}{k}, \\ \mathcal{V}_\gamma(\tau) &= \sum_{k\ge1} \frac{C_{2,\gamma}(k\tau)}{k}. \end{aligned} \tag{E19}$$

All three series converge for every fixed $\gamma > 0$ and $\tau > 0$. With $\langle\cdot\rangle_{1,h}$ denoting expectation in the signal state, the limits are

$$h \log \mathcal{Z}_{1,h} \longrightarrow \mathcal{L}_\gamma(\tau), \tag{E20}$$

$$h\beta\langle \mathcal{H}_{1,h}\rangle_{1,h} \longrightarrow \mathcal{E}_\gamma(\tau),$$

$$h\beta^2 \operatorname{Var}_{1,h}(\mathcal{H}_{1,h}) \longrightarrow \mathcal{V}_\gamma(\tau), \tag{E21}$$

$$hS(\varrho_{1,h}) \longrightarrow \mathcal{L}_\gamma(\tau) + \mathcal{E}_\gamma(\tau). \tag{E22}$$

For each fixed $r > 0$, $r \ne 1$, the Rényi entropy satisfies

$$hS_r(\varrho_{1,h}) \longrightarrow \frac{\mathcal{L}_\gamma(r\tau) - r\mathcal{L}_\gamma(\tau)}{1-r}. \tag{E23}$$

The same statements hold for the collapsed physical ensemble.

Here are the convergent expansions that prove these claims. Each physical normal mode of energy ω has a geometric occupation of mean $(\mathrm{e}^{\beta\omega}-1)^{-1}$ and variance $\mathrm{e}^{\beta\omega}/(\mathrm{e}^{\beta\omega}-1)^2$; different normal modes are independent by the product state. Expanding the geometric series and using nonnegative sums gives

$$\begin{aligned} \log \mathcal{Z}_{1,h} &= \sum_{k\ge1} \frac{1}{k} \operatorname{Tr} \Theta_{1,h}^k, \\ \beta\langle \mathcal{H}_{1,h}\rangle_{1,h} &= \sum_{k\ge1} \frac{1}{k} \operatorname{Tr}\big[(k\beta L_{1,h}^\gamma)\mathrm{e}^{-k\beta L_{1,h}^\gamma}\big], \\ \beta^2 \operatorname{Var}_{1,h}(\mathcal{H}_{1,h}) &= \sum_{k\ge1} \frac{1}{k} \operatorname{Tr}\big[(k\beta L_{1,h}^\gamma)^2\mathrm{e}^{-k\beta L_{1,h}^\gamma}\big]. \end{aligned} \tag{E24}$$

After multiplication by h, (E7) dominates each term by a constant times $k^{-1-1/\gamma}$, a summable sequence for every $\gamma > 0$. The weighted heat theorem therefore applies term by term, proving the first three limits. Finiteness at each h and the exact Gibbs identity $S = \log \mathcal{Z} + \beta\langle\mathcal{H}\rangle$ prove the entropy limit. Similarly $\operatorname{Tr} \varrho_{1,h}^r = \mathcal{Z}_{1,h}(r\beta)/\mathcal{Z}_{1,h}(\beta)^r$ proves (E23). This argument does not differentiate an uncontrolled asymptotic remainder or invoke entropy continuity in unbounded dimension.

Occupation laws require the additional restriction $0 < \gamma < 1$. In this range define

$$\begin{aligned} \mathsf{n}_1 = \mathcal{N}_\gamma(\tau) &= \sum_{k\ge1} C_\gamma(k\tau), \\ \mathsf{n}_Q = \mathcal{N}_{Q,\gamma}(\tau) &= \sum_{k\ge1} C_{Q,\gamma}(k\tau) > 0. \end{aligned} \tag{E25}$$

Because $N_{1,h} = \sum_{k\ge1} \Theta_{1,h}^k$ in trace norm, the heat bound now supplies the summable majorant $k^{-1/\gamma}$. Positivity also bounds a Q-inserted trace by the total trace. Hence

$$\begin{aligned} h \operatorname{Tr} N_{1,h} &\longrightarrow \mathsf{n}_1, \\ h \operatorname{Tr}(QN_{1,h}) &\longrightarrow \mathsf{n}_Q, \\ h \operatorname{Tr} N_{0,h} &\longrightarrow \mathsf{n}_0 := \mathcal{N}_\gamma^{\mathrm{col}}(\tau). \end{aligned} \tag{E26}$$

For $\gamma \geq 1$ this dominating series is not summable, so no h^{-1} occupation law is asserted here. This restriction does not affect (E22)–(E23). The cost of a whole preparation, for $0 < \gamma < 1$, is consequently

$$\langle \mathrm{d}\Gamma(I)\rangle_{1,h} = \frac{\mathsf{n}_1 + o(1)}{h}, \qquad \langle \mathcal{H}_{1,h}\rangle_{1,h} = \frac{\mathcal{E}_\gamma(\tau) + o(1)}{\tau h^{1+\gamma}}. \tag{E27}$$

In particular the frequency exponent $\gamma = 1/2$ requires $O(h^{-1})$ mean quanta and $O(h^{-3/2})$ mean excitation energy, with strictly positive leading coefficients. Equilibration, source and reset times, dissipation, and conversion work are additional resources.

Conditioning (E9) on exactly one excitation gives the normalized one-particle heat state. Its exact probability is

$$\begin{aligned} p_{1,h} &= \frac{\operatorname{Tr}\Theta_{1,h}}{\mathcal{Z}_{1,h}}, \\ p_{1,h} &\leq (\operatorname{Tr}\Theta_{1,h})\mathrm{e}^{-\operatorname{Tr}\Theta_{1,h}}, \\ h\log p_{1,h} &\longrightarrow -\mathcal{L}_\gamma(\tau) < 0. \end{aligned} \tag{E28}$$

The inequality follows from $-\log(1-x) \geq x$. For the last limit, $h\operatorname{Tr}\Theta_{1,h} \to C_\gamma(\tau) > 0$ implies $h\log\operatorname{Tr}\Theta_{1,h} \to 0$, while the partition limit is already proved. Thus the full-ensemble receiver avoids an exponentially rare conditioning event by using a different sample, with the extensive cost in (E27).

The reference coefficients can be evaluated explicitly. Put $\sigma = 1/\gamma$ and $d_\gamma = \Gamma(1+\sigma)/(2\pi)$. Integration of the rank-one reference symbol gives $C^{\mathrm{col}}_\gamma(t) = d_\gamma t^{-\sigma}$, $C^{\mathrm{col}}_{1,\gamma} = \sigma C^{\mathrm{col}}_\gamma$ and $C^{\mathrm{col}}_{2,\gamma} = \sigma(\sigma+1)C^{\mathrm{col}}_\gamma$. The corresponding Bose constants are

$$\mathcal{L}^{\mathrm{col}}_\gamma = d_\gamma\zeta(1+\sigma)\tau^{-\sigma}, \tag{E29}$$

$$\mathcal{E}^{\mathrm{col}}_\gamma = \sigma d_\gamma\zeta(1+\sigma)\tau^{-\sigma},$$

$$\mathcal{V}^{\mathrm{col}}_\gamma = \sigma(\sigma+1)d_\gamma\zeta(1+\sigma)\tau^{-\sigma}, \tag{E30}$$

$$\mathcal{N}^{\mathrm{col}}_\gamma = d_\gamma\zeta(\sigma)\tau^{-\sigma} \tag{E31}$$

$$(\sigma > 1). \tag{E32}$$

The zeta function here is simply $\zeta(s) = \sum_{k\geq 1} k^{-s}$ on its convergent real range.

The high-resolution endpoints of the geometric theorem, (C7) and (C12), extend to these convergent sums:

$$\frac{\mathcal{L}_\gamma}{\mathcal{L}^{\mathrm{col}}_\gamma}, \quad \frac{\mathcal{E}_\gamma}{\mathcal{E}^{\mathrm{col}}_\gamma}, \quad \frac{\mathcal{V}_\gamma}{\mathcal{V}^{\mathrm{col}}_\gamma} \longrightarrow 1 \qquad (\tau \downarrow 0), \tag{E33}$$

and, for $0 < \gamma < 1$,

$$\frac{\mathcal{N}_\gamma}{\mathcal{N}^{\mathrm{col}}_\gamma} \longrightarrow 1, \qquad \frac{\mathcal{N}_{Q,\gamma}}{\mathcal{N}_\gamma} \longrightarrow 1 - \sum_{j=1}^{M}\alpha_j^2. \tag{E34}$$

To check the interchange in this second, successive limit, the eigenvalues of the transverse matrix in B are at most its trace one. With $\xi = w(s)t^{-1/\gamma}u$, this gives $C_{p,\gamma}(t) \leq c_p t^{-1/\gamma}$ for all $t > 0$, using $x^p\mathrm{e}^{-x} \leq c_p\mathrm{e}^{-x/2}$ if necessary. Thus $\tau^{1/\gamma}C_{p,\gamma}(k\tau)$ is bounded by a constant times $k^{-1/\gamma}$, independently of τ. The already proved hot endpoint for each fixed k and dominated convergence prove (E33); for numbers the same argument is summable exactly when $\gamma < 1$. The Q endpoint has the extra factor $1 - \sum_j \alpha_j^2$, proving (E34). It follows in particular that the von Neumann entropy coefficient and each fixed positive-order Rényi entropy coefficient have ratio one to their collapsed counterparts. For $r \neq 1$ the latter counterpart is

$$d_\gamma\zeta(1+\sigma)\tau^{-\sigma}\frac{r^{-\sigma} - r}{1-r} > 0; \tag{E35}$$

the separate $r = 1$ coefficient is $(1+\sigma)d_\gamma\zeta(1+\sigma)\tau^{-\sigma}$.

These are ratios of extensive full-Fock coefficients. They are not the additive one-excitation entropy agreement. Indeed the strict geometric partition deficit (B9) implies, term by term,

$$\begin{aligned} \Delta_\gamma(\tau) &:= \mathcal{L}^{\mathrm{col}}_\gamma(\tau) - \mathcal{L}_\gamma(\tau) > 0, \\ \frac{\mathcal{Z}_{1,h}}{\mathcal{Z}_{0,h}} &= \exp\left[-\frac{\Delta_\gamma(\tau) + o(1)}{h}\right] \longrightarrow 0 \end{aligned} \tag{E36}$$

for every fixed $\tau > 0$ and every nontrivial fixed profile. No arbitrary simultaneous h, τ limit, no moving profile, and no uniform limit in increasing M is included in these conclusions.

4. The efficiency threshold and its Poisson transition

Assume $0 < \gamma < 1$ in this subsection. For any efficiency sequence $0 \leq \eta_h \leq 1$, (E13) and $\log(1+x) \leq x$ give

$$\begin{aligned} &\log(1+\eta_h)\operatorname{Tr}(Q\Theta_{1,h}) \\ &\leq -\log p_{\mathrm{no},1}(\eta_h, Q) \leq \eta_h\operatorname{Tr}(QN_{1,h}). \end{aligned} \tag{E37}$$

Since $\log(1+\eta_h) \geq \eta_h/2$, the heat and number laws imply

$$\begin{gathered} p_{\mathrm{no},1}(\eta_h, Q) \to 0 \\ \Longleftrightarrow \\ \eta_h/h \to \infty, \\ \eta_h/h \to 0 \\ \Longrightarrow \\ p_{\mathrm{no},1} \to 1. \end{gathered} \tag{E38}$$

For necessity in the equivalence, failure of $\eta_h/h \to \infty$ produces a subsequence with bounded η_h/h, on which the upper bound in (E37) keeps the no-click probability bounded away from zero. This is an exact consistency criterion for this specified vacuum-reference receiver, not an optimality theorem for all measurements.

If $\eta_h/h \to \kappa \in [0,\infty)$, let $X_h = \eta_h QN_{1,h}Q$. Then $\operatorname{Tr}X_h \to \kappa\mathsf{n}_Q$ and $\|X_h\| = O(h^{1-\gamma}) \to 0$. Spectral calculus and the scalar bound $0 \leq x - \log(1+x) \leq x^2/2$ give

$$0 \leq \operatorname{Tr}X_h - \operatorname{Tr}\log(I+X_h) \leq \tfrac{1}{2}\|X_h\|\operatorname{Tr}X_h \longrightarrow 0. \tag{E39}$$

Apply this bound to $(1-z)X_h$ in (E11). The generating function tends to $\exp[-(1-z)\kappa\mathsf{n}_Q]$ on $0 \le z \le 1$. Equivalently, evaluating at $z = \mathrm{e}^{-t}$ gives convergence of the nonnegative-variable Laplace transforms to a transform continuous at $t = 0$. Hence

$$K_h \implies \mathrm{Poisson}(\kappa\mathsf{n}_Q), \qquad p_{\mathrm{no},1}(\eta_h, Q) \longrightarrow \mathrm{e}^{-\kappa\mathsf{n}_Q}. \tag{E40}$$

The Poisson approximation is valid in this vanishing-efficiency regime. It is not used for the highly occupied modes of a fixed-efficiency thermal sample.

5. Why an any-click decision is fragile

Suppose a whole detection gate has an independent dark-click probability d_h, common to the hypotheses. Write p_1, p_0 for their no-click probabilities before this background. The equal-prior any-click error is exactly

$$P_{\mathrm{err}} = \tfrac{1}{2}\{1 - (1-d_h)p_0 + (1-d_h)p_1\}. \tag{E41}$$

For the ideal reference $p_0 = 1$, the limiting floor is $d_h/2$ when $p_1 \to 0$. If a fixed $d < 1$ applies independently to each of k independent preparations, the rule declaring signal only when every gate clicks has instead

$$P_{\mathrm{err}}^{(k)} = \tfrac{1}{2}\{d^k + 1 - [1-(1-d)p_1]^k\} \le \tfrac{1}{2}\{d^k + k(1-d)p_1\}. \tag{E42}$$

Thus a finite k suffices for each fixed target error when p_1 is small enough; it pays k preparation costs. When D_h independent channels each have dark-click probability δ_h, $d_h = 1-(1-\delta_h)^{D_h}$. For $D_h = O(h^{-1})$, $\delta_h = o(h)$ suffices for a vanishing gate-level dark floor. If instead $D_h \asymp h^{-1}$ and δ_h is a fixed positive constant below one, $1-d_h$ is exponentially small in h^{-1} and making d_h^k small can itself require exponentially many gates. An aggregated detector has its own gate-level dark probability; the independent-channel formula applies only when that is the actual model.

For a passive unitary sorter V, the detected projection is $E = V^*QV$. If $\theta_h = \|QVP\|$, the reference covariance satisfies

$$\begin{aligned} \mathrm{Tr}(EN_{0,h}) = \mathrm{Tr}(PV^*QVPN_{0,h}) \le \theta_h^2 \,\mathrm{Tr}\, N_{0,h}, \\ \mathbb{P}_0(K \ge 1) \le \eta\theta_h^2 \,\mathrm{Tr}\, N_{0,h}. \end{aligned} \tag{E43}$$

The last step is Markov's inequality. For $0 < \gamma < 1$, keeping this sufficient upper bound below a chosen small fixed δ requires an amplitude budget of order $\theta_h^2 = O(\delta h/\eta)$, with the constant chosen from n_0. This is a worst-case sufficient bound, not a necessary bound for each sorter. The danger from fixed mixing is real: a fixed rotation from v into one fixed unit vector of $Q\mathbb{C}^M$ at every tangential coordinate produces the reference output covariance $\theta^2 N_{0,h}$ on the rotated output channel. For $\theta > 0$ fixed, (E13), applied to the physical reference heat with effective efficiency $\eta\theta^2$, makes its no-click probability tend to zero exponentially. Thus an extensive bright reference can click almost surely under fixed coherent leakage.

A finite-energy reference has a separate quantitative budget. Assume it is defined in the same annihilation representation, has a positive trace-class thermal covariance $N_{\mathrm{r},h}$, commutes with Q, and its one-particle energy restricted to Q is at least $\Delta_h I_Q$. For a detected projection $E_h \le Q$ of finite rank D_h, functional calculus gives

$$1 - p_{\mathrm{r}} \le \eta_h \,\mathrm{Tr}(E_h N_{\mathrm{r},h}) \le \frac{\eta_h D_h}{\mathrm{e}^{\beta\Delta_h} - 1}. \tag{E44}$$

Consequently the signal bound and (E41) imply

$$\begin{aligned} P_{\mathrm{err}} \le \frac{1}{2}\Big\{ d_h + \frac{\eta_h D_h}{\mathrm{e}^{\beta\Delta_h} - 1} \\ + \exp[-\log(1+\eta_h)\,\mathrm{Tr}(E_h\Theta_{1,h})]\Big\}. \end{aligned} \tag{E45}$$

A spectral gap alone does not make an infinite-dimensional thermal trace finite; existence of this reference and the finite detector rank are explicit assumptions. At fixed efficiency and $D_h = O(h^{-1})$, the condition $\beta\Delta_h - \log D_h \to +\infty$ is sufficient for a vanishing reference term. Matching an $\mathrm{e}^{-c/h}$ target by this bound requires a corresponding additional c/h in $\beta\Delta_h$ and an equally controlled dark term. These are requirements of the vacuum-null decision rule. The count threshold below uses a calibrated, possibly nonzero, reference mean.

6. Passive output covariances and a Bose concentration bound

We first justify the statistics after a common imperfect passive receiver. A gauge-invariant thermal state with positive trace-class covariance N has the exact coherent-state representation

$$\varrho_N = \mathbb{E}\,|Z\rangle\langle Z|, \qquad \mathbb{E}Z = 0, \qquad \mathbb{E}[ZZ^*] = N, \tag{E46}$$

where Z is a circular complex Gaussian vector in $\mathfrak{h}$ and $|Z\rangle$ is the normalized coherent vector. This assertion also has an elementary infinite-dimensional justification. In an eigenbasis of N, put $Z_j = \sqrt{n_j}\xi_j$ with independent standard circular complex Gaussians $\mathbb{E}|\xi_j|^2 = 1$. Since $\sum_j n_j < \infty$, Tonelli gives $\sum_j |Z_j|^2 < \infty$ almost surely, so Z lies in $\mathfrak{h}$. The integral of the positive rank-one projectors is a trace-one operator. In one mode, angular integration annihilates off-diagonal number matrix elements and radial integration gives

$$\int_{\mathbb{C}} \frac{\mathrm{e}^{-|z|^2/n}}{\pi n} |z\rangle\langle z|\,\mathrm{d}^2 z = \sum_{m\ge 0} \frac{1}{1+n}\left(\frac{n}{1+n}\right)^m |m\rangle\langle m|. \tag{E47}$$

At $n = 0$ use the vacuum delta measure. Products give the equality on every finite set of modes; increasing finite sets and the finite mean total number identify the full trace-one states. Taking $n_j = \theta_j/(1-\theta_j)$ proves (E46) for (E9).

A contraction $T : \mathfrak{h} \to \mathfrak{g}$, implemented by passive mixing with vacuum ancillas followed by discarding unused

outputs, sends a coherent vector to a coherent vector of amplitude TZ at the measured output. Equation (E46) therefore gives the output covariance TNT^*. With a separate efficiency η, it is

$$C = \eta T N T^*. \tag{E48}$$

For a detection effect $0 \le E \le I$, take $T = \sqrt{E}$. Diagonalizing C is a mathematical step only: total number is invariant under passive output rotations, so the receiver need not know which hypothesis-specific eigenbasis diagonalizes C. Its total count is a sum of independent geometric variables whose means are the eigenvalues c_j of C.

Set $\mu = \operatorname{Tr} C$ and $B_C = \|C\|$. For $|t| \le [4(1+B_C)]^{-1}$, the logarithmic moment generating function is

$$F(t) = -\sum_j \log[1 - c_j(\mathrm{e}^t - 1)],$$
$$F''(t) = \sum_j \frac{c_j(1+c_j)\mathrm{e}^t}{[1-c_j(\mathrm{e}^t-1)]^2}. \tag{E49}$$

For $0 \le t \le [4(1+B_C)]^{-1}$ one has $\mathrm{e}^t \le 4/3$ and $\mathrm{e}^t - 1 \le (4/3)t$; every denominator before squaring is at least $2/3$. Thus $F''(t) \le 3\mu(1+B_C)$. For negative t each denominator is at least one and the same bound follows. These estimates also justify termwise differentiation on compact interior intervals by $\sum_j c_j < \infty$; continuity gives the endpoints. Since $F(0) = 0$ and $F'(0) = \mu$, integration twice proves

$$\log \mathbb{E}\mathrm{e}^{t(K-\mu)} \le 2\mu(1+B_C)t^2, \qquad |t| \le \frac{1}{4(1+B_C)}. \tag{E50}$$

For either tail choose $|t| = u/[4(1+B_C)(\mu+u)]$ in Chernoff's inequality. Substitution gives, for every $u > 0$,

$$\mathbb{P}(K - \mu \ge u),$$
$$\mathbb{P}(K - \mu \le -u) \le \exp\left[-\frac{u^2}{8(1+B_C)(\mu+u)}\right]. \tag{E51}$$

If $\mu = 0$, $K = 0$ almost surely and the same tail bounds hold.

If D is an independent Poisson dark count of known mean ν, then $\log \mathbb{E}\mathrm{e}^{t(D-\nu)} = \nu(\mathrm{e}^t - 1 - t) \le \nu t^2$ for $|t| \le 1/4$. Combining with (E50) proves (E51) for $K + D$, centered at $\mu + \nu$, with μ in the denominator replaced by $\mu + \nu$ and B_C unchanged. This is an independent calibrated Poisson allowance; it is not a claim for arbitrary correlated noise or unknown drift.

7. A count threshold tolerates fixed small effect error

Fix $0 < \gamma < 1$, $\tau > 0$ and $0 < \eta \le 1$. The constants $\mathsf{n}_1, \mathsf{n}_0, \mathsf{n}_Q$ are those in (E26). Suppose the same effect E_h is used under both hypotheses and

$$0 \le E_h \le I,$$
$$\|E_h - Q\| \le e, \tag{E52}$$
$$g := \eta[\mathsf{n}_Q - e(\mathsf{n}_1 + \mathsf{n}_0)] > 0,$$

where e is fixed. The output covariances are $C_{i,h} = \eta\sqrt{E_h} N_{i,h} \sqrt{E_h}$, with means $\mu_{i,h} = \operatorname{Tr} C_{i,h}$. Since $QN_{0,h} = 0$, trace duality gives

$$\mu_{1,h} - \mu_{0,h} \ge \eta\{\operatorname{Tr}(QN_{1,h}) - e[\operatorname{Tr} N_{1,h} + \operatorname{Tr} N_{0,h}]\} \tag{E53}$$
$$= \frac{g + o(1)}{h},$$
$$\mu_{i,h} = O(h^{-1}), \tag{E54}$$
$$\|C_{i,h}\| = O(h^{-\gamma}). \tag{E55}$$

Take the threshold nearest to $(\mu_{1,h} + \mu_{0,h})/2$, declaring signal for counts above it. The two error margins are at least $(\mu_{1,h} - \mu_{0,h})/2 - 1 \ge c_1/h$ for all sufficiently small h. Applying (E51) to both hypotheses yields a constant $c_2 > 0$ such that

$$P_{\mathrm{err}}^{\mathrm{threshold}} \le \exp[-c_2 h^{\gamma-1}]. \tag{E56}$$

Thus the frequency exponent $\gamma = 1/2$ gives $\exp(-c_2/\sqrt{h})$. The occupation norm in (E55) retains Bose bunching; replacing those fluctuations by Poisson ones at fixed efficiency would not prove the stated result.

An independent common Poisson dark count with known mean $\nu_h \le C_\nu/h$ is included by adding ν_h to the threshold and using the preceding concentration result. It changes constants, not the rate. More generally a threshold is valid whenever known background means, or rigorous calibration intervals for them, leave a positive h^{-1} separation and obey the same moment bound. This condition must be checked; arbitrary unknown offsets are not covered. Total-count readout must resolve counts over a range of order h^{-1}; a saturated one-bit detector does not implement this decision.

A reference need not be nearly vacuum for such a threshold. In a finite detected sector of rank at most d/h, a reference occupation bounded by $n_\Delta = (\mathrm{e}^{\beta\Delta_h} - 1)^{-1}$ contributes mean at most $\eta d n_\Delta/h$. It can be included in a calibrated null provided the mean separation and covariance bound remain valid. A sufficiently large fixed $\beta\Delta_h$ may then suffice. On an infinite sector a constant gap without dispersion is not a trace-class thermal state. If instead $H_Q = H_{\mathrm{long}} + \Delta_h I$ with positive confining H_{long}, then the spectral inequality

$$\operatorname{Tr}(\mathrm{e}^{\beta H_Q} - 1)^{-1} \le \frac{\mathrm{e}^{-\beta\Delta_h}}{1 - \mathrm{e}^{-\beta\Delta_h}} \operatorname{Tr} \mathrm{e}^{-\beta H_{\mathrm{long}}} \tag{E57}$$

follows by expanding its geometric series, or by bounding the denominator mode by mode. A controlled $O(h^{-1})$ heat trace thus supplies a finite reference number budget. This is a model of an additive frequency-energy gap, not an assertion that a stiffness penalty adds linearly after taking a square root.

8. One common low-frequency rejection space

The previous rate is limited by the covariance of low-energy Bose modes. We now construct one common filter, designed from the two specified candidate operators, that rejects those modes without discarding the extensive transverse signal. Continue to assume $0 < \gamma < 1$ and fixed $\tau, \eta > 0$.

For a fixed $b > 0$, let $F_{1,h} = \mathbf{1}_{[0,b/h)}(L_{1,h})$. Let $F_{0,h}$ be the analogous projection for L_0 on $P\mathfrak{h}$, extended by zero on Q. By (E3), $\operatorname{rank} F_{i,h} \le Cb/h$. Define

$$\begin{aligned} W_h &= \operatorname{ran} F_{1,h} + \operatorname{ran} F_{0,h} + P \operatorname{ran} F_{1,h}, \\ r_h &= \dim W_h \le C_W b/h, \\ \Pi_h &= I - \operatorname{proj} W_h. \end{aligned} \tag{E58}$$

The space W_h is invariant under P: P maps its first summand into its third, fixes its second, and fixes the image of P. As $P = P^*$, its orthogonal complement is invariant too. Hence $[\Pi_h, P] = [\Pi_h, Q] = 0$ and $\Pi_h \le I - F_{i,h}$ for both candidates. Every vector in its range belongs to the high-energy spectral subspace of each candidate, even though these two spectral projections need not commute. Functional calculus for each one separately gives

$$\Pi_h N_{i,h} \Pi_h \le n_b \Pi_h, \qquad n_b = (\mathrm{e}^{\tau b^\gamma} - 1)^{-1}, \quad i = 0, 1. \tag{E59}$$

For the reference, the additional Q component simply has zero covariance.

For any rank-r orthogonal projection J, Ky Fan's variational inequality and (E8) give

$$\operatorname{Tr}(JN_{1,h}) \le \sum_{j=1}^{r} \nu_j(N_{1,h}) \le \frac{C}{1-\gamma} h^{-\gamma} r^{1-\gamma}. \tag{E60}$$

To see the inequality without any commutation hypothesis, diagonalize N and write $\operatorname{Tr}(JN) = \sum_j \nu_j \langle e_j, Je_j \rangle$; its coefficients lie in $[0, 1]$ and sum to r, so their largest possible weighted sum fills the first r entries. The remaining bound uses $\sum_{j=1}^{r} j^{-\gamma} \le r^{1-\gamma}/(1-\gamma)$. Because $Q(I - \Pi_h)$ is itself an orthogonal projection of rank at most r_h, this implies

$$\begin{aligned} 0 &\le \operatorname{Tr}(QN_{1,h}) - \operatorname{Tr}(Q\Pi_h N_{1,h} \Pi_h) \\ &= \operatorname{Tr}[Q(I - \Pi_h) N_{1,h}] \\ &\le C_* b^{1-\gamma}/h. \end{aligned} \tag{E61}$$

The constant C_* is independent of b and sufficiently small h, with the fixed geometric data, γ and τ understood. Choosing b small therefore preserves a positive fraction of the n_Q/h signal. The choice is not uniform as $\gamma \uparrow 1$.

A finite Fourier realization requires an intersection of spaces, not a product of noncommuting projections. Fix $R > 0$ and let $T_{R,h} = \mathbf{1}_{\{|hD_s| \le R\}} \otimes I_M$. Let $J_{R,h}$ be the orthogonal projection onto the finite-dimensional space $T_{R,h} W_h$, and set

$$\Xi_h = T_{R,h} - J_{R,h}. \tag{E62}$$

The range of $J_{R,h}$ is contained in that of $T_{R,h}$, so Ξ_h is an orthogonal projection. For $x \in \operatorname{ran} T_{R,h}$, $\langle x, T_{R,h} w \rangle = \langle x, w \rangle$ for every $w \in W_h$. Consequently

$$\begin{aligned} \operatorname{ran} \Xi_h &= \operatorname{ran} T_{R,h} \cap W_h^{\perp}, \\ \Xi_h &\le \Pi_h, \\ [\Xi_h, P] &= [\Xi_h, Q] = 0. \end{aligned} \tag{E63}$$

The commutation follows because $T_{R,h}$ commutes with P and W_h is P-invariant, hence so is $T_{R,h} W_h$ and its orthogonal projection. In particular

$$\operatorname{rank} \Xi_h \le \frac{M \ell_\Gamma R}{\pi h} + O(1), \qquad \Xi_h N_{i,h} \Xi_h \le n_b \Xi_h. \tag{E64}$$

No commutation of $T_{R,h}$ with Π_h or of Ξ_h with either Hamiltonian was used.

Since $I - \Xi_h = (I - T_{R,h}) + J_{R,h}$ and all these projections commute with Q, the exact loss of Q-number splits into two nonnegative terms. The rank of $J_{R,h}$ is at most r_h, so

$$\begin{aligned} &\operatorname{Tr}(Q \Xi_h N_{1,h} \Xi_h) \\ &\ge \operatorname{Tr}(QN_{1,h}) - \operatorname{Tr}[Q(I - T_{R,h}) N_{1,h}] - C_* b^{1-\gamma}/h. \end{aligned} \tag{E65}$$

The remaining high-frequency occupation tail is controlled on the actual geometry. Expand $N_{1,h} = \sum_{k \ge 1} \Theta_{1,h}^k$, apply the Fourier insertion law (A33) at each fixed $k\tau$, and then use (E4) with the summable $k^{-1/\gamma}$ majorant. This proves

$$h \operatorname{Tr}[Q(I - T_{R,h}) N_{1,h}] \longrightarrow \mathcal{T}_Q(R),$$

$$\mathcal{T}_Q(R) = \frac{1}{2\pi} \int_\Gamma \int_{|\xi| > R} \sum_{k \ge 1} \operatorname{tr}(Q \mathrm{e}^{-k\tau B(s,\xi)^{-\gamma}}) \, \mathrm{d}\xi \, \mathrm{d}s \longrightarrow 0 \tag{E66}$$

$$(R \to \infty). \tag{E67}$$

The final limit follows from positivity and finiteness of the full integral n_Q. For example choose $b > 0$ with $C_* b^{1-\gamma} < \mathsf{n}_Q/8$ and then R with $\mathcal{T}_Q(R) < \mathsf{n}_Q/8$. These constants are fixed before h tends to zero. Equations (E26) and (E65) then imply, for all sufficiently small h,

$$\operatorname{Tr}(Q \Xi_h N_{1,h} \Xi_h) \ge \frac{\mathsf{n}_Q}{2h}, \qquad \operatorname{Tr}(Q \Xi_h N_{0,h} \Xi_h) = 0. \tag{E68}$$

This proves a finite common receiver with bounded covariance and a strictly extensive retained signal.

Suppose its actual effect has exact rejected-space exclusion and fixed small error inside the retained space:

$$\begin{aligned} 0 &\le E_h \le \Xi_h, \\ \|E_h - \Xi_h Q\| &\le e, \\ e(\mathsf{n}_1 + \mathsf{n}_0) &< \mathsf{n}_Q/4. \end{aligned} \tag{E69}$$

The inequality $0 \le E_h \le \Xi_h$ implies $E_h = \Xi_h E_h \Xi_h$ and $\sqrt{E_h} = \Xi_h \sqrt{E_h}$. Thus

$$C_{i,h} = \eta \sqrt{E_h} N_{i,h} \sqrt{E_h} \le \eta n_b E_h, \qquad \|C_{i,h}\| \le \eta n_b. \tag{E70}$$

Trace duality, (E68) and (E26) leave a positive gap of order h^{-1} between the two means. Both means are $O(h^{-1})$. The midpoint threshold and (E51) now give

$$P_{\rm err}^{\rm filtered} \le \mathrm{e}^{-c/h} \tag{E71}$$

for some $c > 0$ and all sufficiently small h, also with an independent calibrated Poisson dark mean at most C_ν/h. The exact support condition in (E69) is substantive: a small unrestricted effect error alone could admit bright rejected modes and restore the larger covariance norm. The following bound quantifies a sufficient finite rejection tolerance.

9. Finite amplitude leakage and the rate transition

Let the common passive receiver be a contraction $T_h : \mathfrak{h} \to \mathfrak{g}_h$, where the counted output dimension satisfies $\dim \mathfrak{g}_h \le D_*/h$ for a fixed D_*. This explicit output restriction keeps the finite-channel claim valid even when infinitely many rejected input modes can leak. Separate a fixed $0 < \eta \le 1$ efficiency as before, and assume

$$\begin{aligned} &E_{R,h} = \Xi_h T_h^* T_h \Xi_h, \\ &\|E_{R,h} - \Xi_h Q\| \le e, \\ &\ell_h = \|T_h(I - \Xi_h)\|. \end{aligned} \tag{E72}$$

Set $A = T_h\Xi_h$ and $B = T_h(I - \Xi_h)$. Since $\|A\| \le 1$ and $\|B\| = \ell_h$, the identity $T_h^*T_h - E_{R,h} = A^*B + B^*A + B^*B$ gives

$$\begin{aligned} \|T_h^*T_h - \Xi_h Q\| &\le e + 2\ell_h + \ell_h^2, \\ |\operatorname{Tr}[(T_h^*T_h - E_{R,h})N_{i,h}]| &\le (2\ell_h + \ell_h^2)\operatorname{Tr} N_{i,h}. \end{aligned} \tag{E73}$$

For covariance, the positive-operator inequality $(A + B)N(A+B)^* \le 2ANA^* + 2BNB^*$ follows by expanding $(A - B)N(A - B)^* \ge 0$. The first term is bounded in norm by n_b and the second by $\ell_h^2\|N_{i,h}\|$. Therefore

$$\|T_h N_{i,h} T_h^*\| \le 2n_b + 2\ell_h^2\|N_{i,h}\| \le 2n_b + C\ell_h^2 h^{-\gamma}. \tag{E74}$$

The covariance penalty is quadratic in rejected amplitude, although the worst-case mean change includes linear cross terms.

For a precise sufficient uniform gap assumption, require

$$\limsup_{h\downarrow 0}(e + 2\ell_h + \ell_h^2) < \frac{\mathsf{n}_Q}{4(\mathsf{n}_1 + \mathsf{n}_0)}. \tag{E75}$$

Together with (E68) and the number laws, this implies a fixed positive h^{-1} gap for the means of $C_{i,h} = \eta T_h N_{i,h} T_h^*$, while both means remain $O(h^{-1})$. The same is true after a common independent calibrated Poisson dark count of mean at most C_ν/h. Using (E74) in the common concentration bound therefore proves, for some fixed $c > 0$ and all sufficiently small h,

$$P_{\rm err}^{\rm leaky} \le \exp\left[-\frac{c}{h(1 + \ell_h^2 h^{-\gamma})}\right]. \tag{E76}$$

Constants depend on the fixed profile, geometry, γ, τ, η, the chosen b, R, the gap margin and the dark-count bound. If $e < \mathsf{n}_Q/[4(\mathsf{n}_1 + \mathsf{n}_0)]$ and $\ell_h = O(h^{\gamma/2})$, the gap condition is eventually satisfied and the covariance is uniformly bounded, preserving $\exp(-c/h)$. For $\gamma = 1/2$ this is the sufficient amplitude tolerance $O(h^{1/4})$, equivalently a power leakage $O(h^{1/2})$. Fixed sufficiently small leakage gives the available $\exp(-ch^{\gamma-1})$ guarantee; intermediate leakage scales give the displayed interpolation. This is a sufficient receiver budget, not a necessity theorem for every conceivable detector.

The common space W_h uses both known candidate operators and never depends on which hypothesis is true. Its finite Fourier intersection requires the corresponding spectral data and controlled projection onto $T_{R,h}W_h$. Rank $O(h^{-1})$ does not bound numerical conditioning, settings precision, routing depth, stopband performance, physical mode conversion, transmission, reset time or reservoir count. Moreover a finite detector marginal has covariance $\Xi_h N_{i,h}\Xi_h$. It is not generally the Bose function of a compressed Hamiltonian: compression and the nonlinear function $x \mapsto (\mathrm{e}^{\beta x} - 1)^{-1}$ do not commute. The proof uses covariance compression throughout.

10. Common vacuum, mechanical quantization and operational scope

The preceding laws are exact statements in the declared common number-conserving model. For finitely many real canonical coordinates with $[x_j, p_k] = \mathrm{i}\delta_{jk}$ and a real symmetric positive matrix H, the common annihilators $a = (x + \mathrm{i}p)/\sqrt{2}$ satisfy

$$a^*Ha = \tfrac{1}{2}(x^THx + p^THp - \operatorname{Tr} H). \tag{E77}$$

Expansion proves the identity: the antisymmetric commutator contributes $-\operatorname{Tr} H$ and the symmetric cross terms cancel. It realizes H in both quadratures and conserves the chosen particle number. It is a model specification, not a claim that a geometric boundary operator has available local couplings.

An ordinary mass-normalized mechanical Hamiltonian instead has the form $(p^Tp + x^T\Omega^2 x)/2$; its natural annihilators depend on Ω. Already for one oscillator, let

$$a_\omega = \sqrt{\omega/2}\, x + \frac{\mathrm{i}p}{\sqrt{2\omega}}, \qquad a_{\rm r} = \sqrt{\omega_{\rm r}/2}\, x + \frac{\mathrm{i}p}{\sqrt{2\omega_{\rm r}}}. \tag{E78}$$

Solving for x, p and substituting gives

$$a_\omega = \cosh r\, a_{\rm r} + \sinh r\, a_{\rm r}^*, \qquad r = \tfrac{1}{2}\log(\omega/\omega_{\rm r}). \tag{E79}$$

Inverting this relation shows that the ω-vacuum has $\langle a_{\rm r}^* a_{\rm r}\rangle = \sinh^2 r$ quanta in the fixed reference basis. Its reference-vacuum probability is

$$|\langle 0_{\rm r}|0_\omega\rangle|^2 = \frac{2\sqrt{\omega\omega_{\rm r}}}{\omega + \omega_{\rm r}} = \frac{1}{\cosh r}. \tag{E80}$$

For a direct proof, integrate the normalized ground-state wavefunctions $(\omega/\pi)^{1/4}\mathrm{e}^{-\omega x^2/2}$ and $(\omega_{\mathrm{r}}/\pi)^{1/4}\mathrm{e}^{-\omega_{\mathrm{r}}x^2/2}$; squaring their Gaussian overlap gives the displayed fraction. Sending $\omega \to \infty$ suppresses thermal excitation in its own natural basis but drives this fixed-basis vacuum probability to zero. The conversion mixes creation and annihilation operators and is not a passive rotation of a common annihilation basis.

Thus neither a common geometric L^2 space nor a growing mechanical transverse stiffness establishes the reference-vacuum premise of (E14) or (E44). A proposed platform must specify its canonical variables and common detector modes and either realize the number-conserving model or account for the creation part, noise, work and bandwidth of a Bogoliubov conversion. No infinite-mode unitary equivalence of Fock representations is inferred here. Ordinary displacement sensing, passive mixing of layer coordinates and counting excitations in each candidate's own normal modes are not automatically the same common Q occupation measurement.

Finally (E46) supplies a mathematical programmed realization of any finite output marginal: sample its known classical complex Gaussian amplitudes and prepare the associated coherent states. This construction knows the covariance in advance. It is a faithful encoded test of the stated distributions, not by itself an unknown-specimen sensor or a computational quantum advantage. The geometric theorem, the controlled one-excitation route and the full-Bose receiver establish distinct mathematical access statements. Their native physical implementation still requires a geometry-to-generator approximation over the growing band, a common mode map, and preparation, conversion and readout satisfying the stated resource and noise budgets simultaneously.

Appendix F: An exact finite circular counting certificate

1. Finite statement and the declared ensemble

In the unit Dirichlet disk, compare two circular push-forward masses of radii $r_1 = 99/200$ and $r_2 = 101/200$, each of mass $1/2$, with the mass-one reference circle $r_0 = 1/2$. All are uniform in angle. Use the common number-conserving Bose model with $\gamma = 1/2$, $\beta = 1/20$; equivalently $\varepsilon = 1/100$, $\tau = 1/2$. The reference occupies $p = (1,1)^T/\sqrt{2}$ and its orthogonal layer mode $q = (1,-1)^T/\sqrt{2}$ is vacuum. In every angular sector $50 \le |n| \le 500$, detect the mode $q_{\theta_n} = \cos\theta_n\, q + \sin\theta_n\, p$, where $|\theta_n| \le 1/10$ radians. The same settings apply under both hypotheses, but may vary separately across all sectors. Detection efficiency is $\eta = 7/10$, and independent Poisson dark counts have total gate mean $d \in [0, 1/10]$. Declare the split state exactly when the total recorded occupation K is at least five. Then uniformly over these allowances,

$$\begin{aligned} \Pr_0(K \ge 5) &< 0.0057, \\ \Pr_1(K \le 4) &< 0.00072, \\ P_{\mathrm{err}} = \tfrac{1}{2}[\Pr_0(K \ge 5) + \Pr_1(K \le 4)] &< 0.0032. \end{aligned} \tag{F1}$$

There are 902 measured output modes and 1804 modes in the full two-layer prepared band. This is a statement about the explicitly aligned Bose model or its exact band marginal. It is not a single-phonon error statement or a claim about an unmodified membrane.

2. Exact Green blocks with both angular signs

For $z = re^{i\theta}$ and $z' = se^{i\phi}$ in the disk,

$$G(z,z') = \frac{1}{2\pi}\log\left|\frac{1 - z\overline{z'}}{z - z'}\right|.$$

Expanding $\log|1 - te^{i\psi}| = -\sum_{n\ge1} t^n\cos(n\psi)/n$ for $|t| < 1$ gives

$$\begin{aligned} G = \frac{1}{2\pi}\Big[&-\log\max(r,s) \\ &+ \sum_{n\ge1}\frac{(\min(r,s)/\max(r,s))^n - (rs)^n}{n} \\ &\cos(n(\theta-\phi))\Big]. \end{aligned}$$

For $r = s$ the coefficients follow by the integrable logarithmic limit. Convolution uses the normalized physical angular measure $\mathrm{d}\phi/(2\pi)$; a nonzero complex Fourier coefficient of the cosine is $1/2$. After conjugating the masses, the nth block is

$$(K_n)_{ij} = \frac{\sqrt{\alpha_i\alpha_j}}{4\pi n}\left[\left(\frac{\min(r_i,r_j)}{\max(r_i,r_j)}\right)^n - (r_ir_j)^n\right], \quad n \ge 1. \tag{F2}$$

Thus its prefactor here is $1/(8\pi n)$. The block for $-n$ is identical. The reference nonzero scalar eigenvalue is

$$k_{0,n} = \frac{1 - (1/2)^{2n}}{4\pi n}, \qquad K_{0,n} = k_{0,n}P. \tag{F3}$$

Both total physical masses are one. Positivity of the Green form on distinct layers gives positive blocks; the numerical certificate additionally verifies their smaller eigenvalues are strictly positive throughout this band. There are 451 positive measured integers. The independent $-n$ sectors, equivalently the second real harmonic at each positive n, give multiplicity two. This is not a second copy of the experiment. Rotational invariance makes full angular blocks reducing for both candidates; their layer matrices need not commute with each other.

3. Covariance and sorter bounds

Let

$$g(x) = \frac{1}{e^{\beta/\sqrt{x}} - 1}, \qquad x > 0, \quad N_n = g(K_n).$$

Write $k_- < k_+$ for the eigenvalues of K_n, $n_\pm = g(k_\pm)$, and $b_n = (n_+ - n_-)/(k_+ - k_-) > 0$. The exact two-dimensional functional calculus

$$g(K_n) = \frac{n_+ + n_-}{2} I + b_n \left(K_n - \frac{k_+ + k_-}{2} I \right)$$

gives

$$\begin{aligned} q_\theta^* N_n q_\theta = \frac{n_+ + n_-}{2} \\ + b_n \big[\frac{K_{n,11} - K_{n,22}}{2} \sin(2\theta) \\ - K_{n,12} \cos(2\theta) \big]. \end{aligned} \tag{F4}$$

Since $K_{n,12} > 0$ and g is increasing, for $a = 1/10$ the detected target mean is bounded below, uniformly over $|\theta| \le a$, by

$$\begin{aligned} \ell_n = \eta \big[\frac{n_+ + n_-}{2} - b_n K_{n,12} \\ - \frac{|b_n (K_{n,11} - K_{n,22})|}{2} \sin(2a) \big] > 0. \end{aligned} \tag{F5}$$

The inequality uses $\cos(2\theta) \le 1$ in the negative term and $|\sin(2\theta)| \le \sin(2a)$. These separate worst-case bounds need not be simultaneously attained. Under the reference the detected mean is exactly $\eta g(k_{0,n}) \sin^2\theta$, and therefore at most

$$u_n = \eta g(k_{0,n}) \sin^2 a. \tag{F6}$$

4. Geometric occupation tails and interval arithmetic

A single mode in a gauge-invariant Gaussian state with mean ν has geometric occupation distribution $\Pr(X = k) = \nu^k/(1+\nu)^{k+1}$, so

$$\mathbb{E} z^X = [1 + \nu(1 - z)]^{-1}.$$

One direct verification uses its positive coherent-state mixture: draw a circular complex Gaussian amplitude with variance ν, then average the coherent-state Poisson generating function $\exp[-(1-z)|\zeta|^2]$. Loss of efficiency η rescales the variance by η. Distinct angular sectors are independent conditional on the fixed sorter settings; no independence of the two layer eigenmodes after sorting is assumed or needed. The same count structure is classical thermal optics [10, 13].

For null tilt $t = 3/2$, each denominator $1 - u_n(e^t - 1)$ is positive. Markov's inequality, monotonicity in the geometric mean and the Poisson generating function imply

$$\begin{aligned} B_0 = \exp\big[-5t - 2 \sum_{n=50}^{500} \\ \log(1 - u_n(e^t - 1)) \\ + \tfrac{1}{10}(e^t - 1) \big] \\ \ge \Pr_0(K \ge 5). \end{aligned} \tag{F7}$$

For the miss event $K \le 4$, choose $s = 1$. Dropping nonnegative independent dark counts increases its probability. Its decreasing Laplace transform and (F5) give

$$\begin{aligned} B_1 = \exp\big[4s - 2 \sum_{n=50}^{500} \\ \log(1 + \ell_n(1 - e^{-s})) \big] \\ \ge \Pr_1(K \le 4). \end{aligned} \tag{F8}$$

The coefficient 4, rather than 5, is the largest integer in the miss event. These bounds hold conditional on every allowed collection of angles, so they also hold after any classical randomization of the angles independent of the prepared state, even if the angles are correlated across sectors.

The accompanying checker evaluates these finite expressions with Arb ball arithmetic [25]. Every input rational is formed by exact integer division; π, roots, exponentials, sines and logarithms receive interval enclosures. For every n it checks $k_- > 0$, $k_+ - k_- > 0$, $b_n > 0$, $K_{n,12} > 0$, $\ell_n > 0$, and $1 - u_n(e^t - 1) > 0$. Explicitly, the discriminant is computed as

$$\sqrt{(K_{n,11} - K_{n,22})(K_{n,11} - K_{n,22}) + 4K_{n,12}K_{n,12}},$$

so a tiny zero-straddling diagonal difference is handled by valid interval multiplication. Finally it proves, as strict interval inequalities against exact rationals,

$$B_0 < \frac{57}{10000}, \qquad B_1 < \frac{72}{100000}, \qquad \frac{B_0 + B_1}{2} < \frac{32}{10000}.$$

At 256-bit working precision the enclosures lie inside

$$\begin{aligned} B_0 \in (0.00566995032088188856, \\ 0.00566995032088188858), \\ B_1 \in (0.00071698918995924630, \\ 0.00071698918995924633), \\ (B_0 + B_1)/2 \in (0.00319346975542056743, \\ 0.00319346975542056746). \end{aligned}$$

These finite rational comparisons prove (F1). Higher-precision replay checks the same expressions. The quantities B_0, B_1 are upper bounds on tails, not exact tail probabilities or Monte Carlo confidence limits.

5. Resource values and exact scope of the marginal

The full prepared band occupation and excitation-energy sums are

$$\overline{N}_1 = 2\sum_{n=50}^{500}(n_- + n_+),$$
$$\overline{E}_1 = 2\sum_{n=50}^{500}(n_-/\sqrt{k_-} + n_+/\sqrt{k_+}),$$
$$\overline{N}_0 = 2\sum_{n=50}^{500} g(k_{0,n}),$$
$$\overline{E}_0 = 2\sum_{n=50}^{500} g(k_{0,n})/\sqrt{k_{0,n}}.$$

Their interval-enclosed values, rounded here for interpretation, are

$$(\overline{N}_1, \overline{N}_0) \simeq (61.79306, 81.31572),$$
$$(\overline{E}_1, \overline{E}_0) \simeq (3550.20358, 3590.80726).$$

These are pre-detection band resources and omit zero-point energy. In contrast, the uniform detected target mean lower bound is about 18.00837, and the reference upper bound before dark counts is about 0.56732. A complete equilibrium sample has additional occupation and energy outside the band. The unequal prepared occupations are not equal thermal scalar data; this finite point certifies access and discrimination, not the asymptotic thermal-scalar comparison or an advantage over every receiver at matched brightness.

The full two-layer angular band is an exact finite reducing subspace. Within each block the detected rank-one output need not reduce the Hamiltonian. Its marginal is nevertheless Gaussian with the exact compressed covariance used in (F4). Tracing out the remaining modes does not assert that they are unoccupied and introduces no spectral-tail approximation into this measurement. One can prepare the specified band marginal alone, for example by sampling correlated classical complex Gaussian amplitudes with its covariance. This is a programmed simulation with known covariance. In general the covariance V^*NV must not be replaced by $(e^{\beta V^* HV} - I)^{-1}$, because compression and functional calculus do not commute. Reproducing this marginal is not global Fock-state approximation in trace distance.

The finite certificate assumes ideal exclusion of angular modes outside $50 \le |n| \le 500$, the stated efficiency throughout the band, real within-sector rotation errors only, number resolution through the threshold, and the independent gate-level Poisson background. It gives no finite numerical allowance for stopband leakage, cross-sector mixing, phase errors, finite radial mass width, changing carrier stiffness, or bulk inertia. The general asymptotic rejection theorem does not by itself supply those finite tolerances. The common vacuum and annihilation modes are declared preparation assumptions. The classical mode-projection receiver and its crosstalk sensitivity have direct precedents [10, 11]; the result here is this exact geometric finite instance with its resource and tolerance ledger.

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